Water Quality Sampling Data Sheet:

Date: 5-27-14
Fishpond: Heeia / Kelii's Curly
Initial Observations:

Start Time	9:50am	% Cloud Cover	75-80%
Finish Time	1:30pm		
Tide	low @ 7:~~10~~am -.33 ft	Precipitation	None
Air Temp	76 °F	Other Comments:	Rained hard over weekend
Wind Speed/Direction	ENE 10-15kt		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)		46cm	49		68				39	46
Surf. Water Temp °C		24.29°c	25.27		25.63				25.64	25.58
Surf. Salinity		15.51	9.28		21.92				31.97	24.86
Surf. DO %		100.7	92.8		89.9				110.0	98.5
Surf. DO mg/L		7.71	7.17		6.47				7.48	6.98
Surf. pH		8.01	7.67		8.01				8.19	8.06
Bottom Temp °C		26.52	27.98		26.66				25.66	26.13
Bottom Salinity		19.50	27.46		28.52				31.96	29.31
Bottom DO%		94.1	72.5		93.5				102.2	114.3
Bottom DO mg/L		6.83	5.04		6.42				7.26	7.73
Bottom pH		8.11	7.87		8.09				8.20	8.17
NO_3^- (Nitrate) mg/L		0.33	0.64		0.65				1.86	0.39
NO_2^- (Nitrite) mg/L		0.000	0.010		0.002				0.008	0.001
NH_4 (Ammonia) mg/L		0.00	0.03		0.02				0.00	0.00
PO_4^{-3} (Phosphate)		.08	0.06		0.00				0.02	0.09
SiO_2 (Silica) mg/L		13	~~0.005~~		9				1	6
Turbidity (FAU)		4	19		5				1	4

- clip board w/ rubber band
- cloth/rag/towel

Water Quality Sampling Data Sheet:

Date: 6/5
Fishpond: Heeia, Keki

Initial Observations:

Start Time	11.20	% Cloud Cover	80-90% but thin, so pretty sunny
Finish Time	~~12:30~~ 5pm		
Tide		Precipitation	off and on trade wind showers
Air Temp	80	Other Comments:	
Wind Speed/Direction	10-15 ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	58	59	61	61	91	53	47	49	56	58
Surf. Water Temp °C	25.47	24.97	25.37	~~26.50~~	26.24	25.89	26.81	26.34	26.21	26.46
Surf. Salinity	19.59	18.01	14.01	~~23.27~~	24.57	29.70	22.68	25.57	28.18	27.44
Surf. DO %	106.0	109.5	96.1	164.8	112.2	138.8	110.2	110.0	116.3	130.5
Surf. DO mg/L	7.78	8.20	7.27	7.40	7.88	9.65	7.77	7.68	8.01	8.99
Surf. pH	8.20	8.26	7.80	8.32	8.32	8.44	8.37	8.34	8.43	8.49
Bottom Temp °C	27.22	25.01	27.45	26.67	26.35	26.01	26.91	27.18	26.31	26.49
Bottom Salinity	24.26	21.9	24.71	24.32	26.63	28.71	22.93	26.44	28.15	27.87
Bottom DO%	129.1	106.6	97.6	102.7	150.2	141.3	112.9	116.6	118.0	131.6
Bottom DO mg/L	8.94	9.08	6.78	7.17	10.41	9.64	7.92	8.98	8.06	9.02
Bottom pH	8.66	8.41	8.26	8.35	8.52	8.51	8.36	8.44	8.44	8.50
NO_3^- (Nitrate) mg/L	0.5	0.9	0.7	0.7	0.75	1.05	0.79	0.89	0.84	0.58
NO_2^- (Nitrate) mg/L	0.00	0.007	0.4	0.005	~~0.005~~	0.005	0.005	0.004	0.005	0.008
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.02	0.01	0.02	0.02	0.01	0.00	0.00
PO_4^{-3} (Phosphate)	0.00	0.09	0.18	0.05	0.08	0.09	~~0.03~~	0.01	0.09	2.35
SiO_2 (Silica) mg/L	7	11	13	9	7	3	7	6	3	4
Turbidity (FAU)	9	6	13	0	20	4	5	7	0	0

sample was purple!

didn't use saltwater nitrate, used freshwater

Water Quality Sampling Data Sheet:

Date: 6/19
Fishpond: Heeia

Initial Observations:

Start Time	9:30a	% Cloud Cover	15-20% sunny
Finish Time	10:45a		
Tide		Precipitation	none today morning showers recently
Air Temp	80°F	Other Comments:	bright sunny summer days
Wind Speed/Direction	NE 5-10		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	51	61	67	64	93	63	55	59	59	66
Surf. Water Temp °C	25.16	24.69	25.50	25.74	25.84	26.02	26.39	25.95	26.24	26.54
Surf. Salinity	23.18	24.21	19.60	24.77	28.29	31.25	26.91	30.01	32.37	30.42
Surf. DO %	109.4	106.5	97.8	89.9	127.9	122.2	97.3	102.7	130.2	112.0
Surf. DO mg/L	7.90	7.70	7.18	6.31	8.84	8.27	6.72	7.52	8.75	7.60
Surf. pH	8.37	8.43	8.19	8.25	8.53	8.48	8.40	8.38	8.45	8.43
Bottom Temp °C	25.32	25.96	27.29	26.87	25.81	26.15	26.55	26.08	26.27	26.52
Bottom Salinity	104.5	27.09	29.30	28.31	29.25	31.69	27.09	31.02	33.51	30.45
Bottom DO%	103.9	114.5	103.7	94.4	138.2	121.5	135.0	106.3	110.5	106.7
Bottom DO mg/L	7.61	8.68	6.99	6.14	9.55	8.04	9.35	7.04	7.40	7.21
Bottom pH	8.38	8.55	8.37	8.39	8.56	8.46	8.46	8.34	8.36	8.41
NO_3^- (Nitrate) mg/L	0.71	0.47	0.81	0.60	0.81	1.59	1.11	1.18	1.57	1.46
NO_2^- (Nitrate) mg/L	0.007	0.006	0.008	0.000	0.003	0.012	0.001	0.000	0.003	0.004
NH_4 (Ammonia) mg/L	0.00	0.00	0.03	0.00	0.01	0.00	0.00	0.00	0.01	X wrong!
PO_4^{-3} (Phosphate)	0.15	0.09	0.07	0.20	0.10	0.08	0.07	0.02	0.00	0.10
SiO_2 (Silica) mg/L	5	5	10	8	7	3	4	3	3	4 okay
Turbidity (FAU)	5	10	14	9	5	5	10	4	0	5

5 samples

Water Quality Sampling Data Sheet:

Date: June 26 & 27 – 5 samples

Fishpond: Heeia

Initial Observations:

Start Time	7:40a 6/26	% Cloud Cover	90 %
Finish Time	3pm 6/26		
Tide	extreme low outside	Precipitation	showers moving on shore
Air Temp		Other Comments:	rained last night
Wind Speed/Direction	<-10 ENE		

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Site #	1	2	3	4	~~5~~ X	6	~~7~~ X	8	9	10
Depth (cm)	38	56	59	51	38	41	70	37	48	46
Surf. Water Temp °C	25.21	25.60	26.02	25.95	25.74	25.43	25.60	25.18	24.70	25.40
Surf. Salinity	28.89	30.47	24.04	29.35	25.68	30.50	28.67	27.71	27.66	29.21
Surf. DO %	58.2	48 5	65.5	65.4	64.0	95.0	68.0	87.7	98.4	67.2
Surf. DO mg/L	4.06	3.52	4.54	4.48	4.37	6.56	4 75	6.17	7.02	4.68
Surf. pH	7.83	7.96	7.88	7.95	7.98	~~7.96~~ 8.07	8 01	8.04	8 11	8 02
Bottom Temp °C	25.36	25.62	26 05	26.01	25.94	25.46	26.11	25.23	25,31	25.63
Bottom Salinity	29 06	30.48	28.15	29.36	28.75	30.58	30.05	28.18	28 22	29.49
Bottom DO%	55.8	45.3	78.4	63 4	58 5	94.1	58.2	79.3	95.7	63 8
Bottom DO mg/L	3.88	3 14	5.44	4.39	4.03	6.47	3 91	5.80	6.74	4.59
Bottom pH	7.87	7 95	7.84	7.97	7.96	8 14	7.96	8 00	8.10	8 01
NO_3^- (Nitrate) mg/L	0.78	2.34	0.31	0.54	0.80	0.55	0.66	2.53	0.55	0.65
NO_2^- (Nitrite) mg/L	0.006	0.006	0.000	0.000	0.007	0.002	0.007	0.003	0.000	0.007
NH_4 (Ammonia) mg/L	0.03	0.02	0.05	0.06	0.02	0.01	0.00	0.05	0.03	0.01
PO_4^{-3} (Phosphate)	0.04	0.06	0.10	0.14	0.03	0.02	2.13	0.00	0.12	0.07
SiO_2 (Silica) mg/L	4	~~11~~	11	5	5	3	5	5	6	4
Turbidity (FAU)	4	0	14	16	18	3	7	2	3	7

sample was purple

Water Quality Sampling Data Sheet:

Date 7/2/14
Fishpond Hecia

Initial Observations:

Start Time	8 30	% Cloud Cover	70%
Finish Time			
Tide	LOW	Precipitation	No much / Light
Air Temp		Other Comments:	normal tradewind weather
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	45	56	60	61	91	50	56	47	52	54
Surf. Water Temp °C	26.53	25.49	25.03	26.62	25.89	26.28	2683	26 57	26.5	27.13
Surf. Salinity	25.34	2250	27.67	24.5	18.36	29.44	24.27	28.30	29.57	29.51
Surf. DO %	82.8	89	76 0	90 1	88.8	99.3	102.1	101 3	101.3	91.1
Surf. DO mg/L	5 77	6.60	6.01	6 32	6.54	6.77	7 0	6 95	6.90	6.13
Surf. pH	8 05	8.12	7.58	8 17	8.01	8.26	8 20	8.23	8 29	8.22
Bottom Temp °C	26.53	25.76	28.8	26.92	28.90	26.57	27 35	26 61	26 52	27.22
Bottom Salinity	25 14	23.37	27.6	25 53	30.50	29.46	24.82	28 30	29 59	29.50
Bottom DO%	86 4	91.4	46 8	87.3	45.9	101.9	105.0	98 8	101 2	91.6
Bottom DO mg/L	6 01	6.47	3 15	6 04	2.83	6.86	7 08	6 77	6.88	6.05
Bottom pH	8 21	8.22	7 84	8 18	8.00	8.27	8 23	8 23	8 29	8.21
NO_3^- (Nitrate) mg/L	0 94	0.67	0.00	0.83	0.80	0.84	1.13	0.73	1.03	0.04
NO_2^- (Nitrite) mg/L	0.007	0 23	0.00	0.003	0.009	0.001	0.002	0.003	0.004	0.001
NH_3 (Ammonia) mg/L	0.03	0.07	0.05	0.05	0.08	0.03	0.03	0.00	0.02	0.00
PO_4^{-3} (Phosphate)	0.14	0.06	0.13	0.08	0.15	0.34	0.08	0.00	0.00	0.04
SiO_2 (Silica) mg/L	7	12	16	6	13	3	7	5	4	5
Turbidity (FAU)	5	2	49	2	8	2	4	0	3	7

Water Quality Sampling Data Sheet:

Date: 9/7

Fishpond: Heeia

Initial Observations:

Start Time	3:20	% Cloud Cover	80%
Finish Time	8/9 8:45 am		
Tide	High	Precipitation	Little Bit / Morning showers
Air Temp		Other Comments:	Finish collecting: 4.30
Wind Speed/Direction	ENE		

Site #	1	2	3	5	4	6	7	8	9	10
Depth (cm)	64	72	78	89	80	63	58	63	64	70
Surf. Water Temp °C	26.8	26.41	26.36	26.61	26.63	26.92	27.78	27.32	27.11	26.77
Surf. Salinity	25.71	22.77	18.03	23.56	22.64	25.33	29.82	27.08	29.24	26.26
Surf. DO %	119.8	98.2	92.2	108.9	105.5	113.0	164.5	124.1	121.81	135.1
Surf. DO mg/L	8.36	6.91	6.7	7.62	7.48	7.81	10.95	8.31	8.00	8.88
Surf. pH	8.35	8.29	8.02	8.31	8.31	8.38	8.45	8.41	8.40	8.48
Bottom Temp °C	26.89	26.72	27.03	27.02	27.19	28.08	28.12	27.81	28.10	28.25
Bottom Salinity	25.78	27.73	26.74	27.49	25.73	32.98	32.01	33.51	33.80	33.75
Bottom DO%	124.7	112.5	114.7	139.4	120.3	184.9	166.0	160.9	189.7	196.3
Bottom DO mg/L	8.57	7.71	7.86	9.55	8.14	12.08	10.83	10.80	12.85	13.10
Bottom pH	8.39	8.36	8.35	8.47	8.40	8.65	8.46	8.51	8.59	8.60
NO_3^- (Nitrate) mg/L	0.40	0.26	1.41	0.66	0.75	0.98	0.18	0.74	0.15	0.64
NO_2^- (Nitrate) mg/L	0.001	0.006	0.002	0.003	0.000	0.002	0.007	0.003	0.001	0.003
NH_4 (Ammonia) mg/L	0.04	0.01	0.02	0.00	0.06	0.00	0.02	0.04	0.00	0.00
PO_4^{-3} (Phosphate)	0.08	0.08	0.17	0.04	0.19	0.06	0.07	0.11	0.08	0.19
SiO_2 (Silica) mg/L	3	4	5	5	3	5	7	4	3	3
Turbidity (FAU)	0	3	2	1	0	1	2	2	1	2

Water Quality Sampling Data Sheet:

Date: 7/9/2014
Fishpond: Heeia

Initial Observations:

Start Time	1130a	% Cloud Cover	60%
Finish Time	1245p		
Tide	falling + high	Precipitation	
Air Temp		Other Comments:	
Wind Speed/Direction	NE 5-10 mph		

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Site #	1	2	~~3~~ X	4	~~5~~ X	6	7	8	9	10
Depth (cm)	48	61	96	63	66	79	70	73	75	89
Surf. Water Temp °C	26.52	26.38	26.24	26.63	25.90	26.92	27.10	26.86	26.76	27.30
Surf. Salinity	23.74	24.84	23 00	22.25	18.35	34.61	23.88	30 76	34.78	30.7‰
Surf. DO %	151.1	128 8	108.1	92.7	80.2	103.1	89.0	119 9	118.6	119.5
Surf. DO mg/L	10.58	9.11	7.68	6.54	6.23	6.79	6.18	8.03	8.00	7.96
Surf. pH	8 45	8 37	8.26	8 15	7.67	8.29	8.17	8 33	8.31	8.36
Bottom Temp °C	26 50	26 48	26.83	26.65	26.58	26.93	27.20	26.73	26.78	26.82
Bottom Salinity	23.73	24.84	26.30	24.76	24.68	34.65	24.40	31 30	34 81	33.50
Bottom DO%	148 8	143.0	132.9	86.8	132.6	102.4	56.0	115 6	108 2	102.8
Bottom DO mg/L	10 50	9.37	9.22	5.78	9.38	6.75	6.09	7.76	7.07	7.21
Bottom pH	8 47	8 39	8.38	8.18	9.38	8.30	8.18	8 33	8.31	8.32
NO_3^- (Nitrate) mg/L	0.94	1.23	0.108	0.96	0.96	1.56	0.76	1.26	0.64	2.17
NO_2^- (Nitrite) mg/L	0.000	0.007	0.001	0.010	0.003	0.014	0.06	0.004	0.001	0.007
NH_4 (Ammonia) mg/L	0.01	0.00	0.03	0.03	0.03	0.00	0.00	0.01	0.00	0.02
PO_4^{-3} (Phosphate)	0.00	0.11	0.09	0.11	0.01	0.00	0.00	0.08	0.07	0.14
SiO_2 (Silica) mg/L	5	5	12	8	9	2	8	3	1	4
Turbidity (FAU)	6	3	4	5	0	3	13	0	X	0

Water Quality Sampling Data Sheet

Date: 07/14/14
Fishpond: HeleA

Initial Observations:

Start Time	12:45 pm	% Cloud Cover	80%
Finish Time	01:20 pm.		
Tide	Low tide rising	Precipitation	drizzled
Air Temp		Other Comments:	
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	35	51	52	51	85	46	40	44	48	58
Surf. Water Temp °C	28.35	27.9	26.5	27.25	27.28	29.01	28.82	28.85	28.94	29.02
Surf. Salinity	29.39	20.01	7.08	17.40	15.62	20.82	22.85	30.41	32.41	27.49
Surf. DO %	141.5	109.4	76.1	104.7	96.3	149.6	108.8	131.8	142.9	153.3
Surf. DO mg/L	9.64	7.28	5.92	7.9	6.71	9.22	7.29	9.18	9.19	10.03
Surf. pH	8.35	8.17	7.51	8.05	7.87	8.48	8.14	8.5	8.50	8.54
Bottom Temp °C	28.9	29.3	28.52	27.85	28.7	29.06	28.72	29.00	28.77	27.75
Bottom Salinity	25.5	25.62	18.70	19.29	31.21	30.78	25.35	30.48	32.42	29.53
Bottom DO%	179.2	173.4	86.4	96.1	78.0	131.9	91.4	137.1	137.5	141.3
Bottom DO mg/L	11.83	11.73	6.37	6.73	4.83	8.78	6.06	8.80	9.04	9.10
Bottom pH	8.70	8.6	7.82	8.10	8.01	8.47	8.18	8.19	8.49	8.52
NO_3^- (Nitrate) mg/L	1.01	0.70	0.31	0.88	0.29	1.02	0.93	0.31	0.54	0.41
NO_2^- (Nitrite) mg/L	0.002	0.004	0.008	0.002	0.002	0.004	0.007	0.000	0.002	0.007
NH_4 (Ammonia) mg/L	0.02	0.02	0.02	0.01	0.08	0.00	0.04	0.03	0.00	0.02
PO_4^{-3} (Phosphate)	0.07	0.11	0.06	0.10	0.05	0.13	0.09	0.00	0.00	0.00
SiO_2 (Silica) mg/L	10	10	19	13	11	4	9	3	3	0
Turbidity (FAU)	1	5	6	14	12	7	13	0	1	15

Water Quality Sampling Data Sheet:

Date: 07/16/14
Fishpond: 4e'e14

Initial Observations:

Start Time	9:30am	% Cloud Cover	65%
Finish Time	03:45pm		
Tide	Rising	Precipitation	None | Morningshowers
Air Temp		Other Comments:	
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	49	56	70	60	76	50	43	48	57	54
Surf. Water Temp °C	27.15	26 70	26.40	28 12	27 36	27.46	28 .31	27 94	27 53	27.82
Surf. Salinity	23 51	16·70	11.41	24 78	21 50	31.10	24 82	27 20	31 70	30 .29
Surf. DO %	106 1	93.8	78.9	92 1	80 0	115·1	93.7	99.5	99 .2	103.6
Surf. DO mg/L	7 32	6 85	5.92	6·28	5 ·61	7·50	6.40	6 70	6.54	6·88
Surf. pH	8 .30	8·05	7.75	8.15	8 00	8·35	8.17	8 20	8.20	8.24
Bottom Temp °C	29.05	27.72	29 60	28 .38	29.04	27.48	28.36	28·16	27.54	27.88
Bottom Salinity	25 46	28 .22	27.75	26.94	21·87	31.10	24 84	27.66	31.79	30.60
Bottom DO%	110 0	84 9	48 9	92.0	105·2	113.1	93.2	99.1	98.5	104.0
Bottom DO mg/L	7 81	5. 75	3 25	6 10	7 .01	7 51	6 33	6 66	6.51	6·70
Bottom pH	8.58	8 .25	7 ·83	8·20	8 ·28	8 .35	8·18	9 .25	8 .22	8.24
NO_3^- (Nitrate) mg/L	0.92	0 .63	0.21	0 81	0.95	0.22	1.64	0.54	0.38	0.51
NO_2^- (Nitrite) mg/L	0.002	0 .006	0.004	0.005	0.007	0.000	0.015	0.004	0.013	0.003
NH_4 (Ammonia) mg/L	0 .00	0 .00	0 .00	0 .00	0.02	0 .00	0.01	0 .00	0 .00	0 .00
PO_4^{-3} (Phosphate)	0.02	0 .05	0 .09	0 .07	0 .10	0 .07	0 .09	0 .11	0 .04	0 .11
SiO_2 (Silica) mg/L	8	12	13	10	8	2	7	4	3	4
Turbidity (FAU)	1	4	10	2	18	0	12	2	0	0

I don't know if chemical tests were recorded in numerical site order or in order shown - Carly
Kelii said its all correct...

Water Quality Sampling Data Sheet:

Date: 07/21/14
Fishpond: Hakioa

Initial Observations:

Start Time	12:30 pm	% Cloud Cover	75%
Finish Time	01:20 pm		
Tide	High, dropping	Precipitation	drizzling, morning showers
Air Temp		Other Comments:	Huge rain storm on 7/19, brown water advisory all over Oahu 7/20 & 7/21
Wind Speed/Direction			

Site #	1	2	6 X	4	5	3 X	7	8	9	10
Depth (cm)	80	95	86	94	117	98	81	84	92	90
Surf. Water Temp °C	28.18	28.19	29.08	29.64	28.58	28.56	29.47	28.95	29.78	29.55
Surf. Salinity	3.03	1.60	3.27	3.54	2.01	1.48	4.00	8.70	3.31	2.07
Surf. DO %	93.2	76.3	66.7	79.1	73.6	71.5	78.5	80.6	75.6	86.1
Surf. DO mg/L	6.80	5.82	4.89	5.79	5.68	5.46	5.96	5.31	5.48	6.24
Surf. pH	7.82	7.82	7.76	7.70	7.47	7.53	7.81	7.93	7.84	7.76
Bottom Temp °C	27.31	27.59	28.03	27.83	27.92	27.51	28.08	28.14	28.02	27.81
Bottom Salinity	25.32	25.32	25.66	24.52	25.65	22.75	24.34	26.43	27.47	25.71
Bottom DO%	84.9	93.2	103.0	72.8	87.9	68.4	96.6	93.3	113.9	103.1
Bottom DO mg/L	5.91	6.38	6.84	4.99	5.98	4.42	6.61	6.30	7.77	6.96
Bottom pH	8.20	8.21	8.22	7.84	8.01	7.50	8.12	8.14	8.27	8.11
NO_3^- (Nitrate) mg/L	0.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
NO_2^- (Nitrite) mg/L	0.003	0.008	0.009	0.000	0.000	0.000	0.001	0.008	0.004	0.003
NH_4 (Ammonia) mg/L	0.21	0.13	0.24	0.25	0.14	0.10	0.21	0.17	0.20	0.14
PO_4^{-3} (Phosphate)	0.25	0.31	0.13	0.15	0.16	0.11	0.11	0.15	0.17	0.00
SiO_2 (Silica) mg/L	12	7	9	6	13	8	11	8	7	13
Turbidity (FAU)	24	20	28	31	38	35	27	26	29	31

Keahi

Water Quality Sampling Data Sheet:

5 pts
7/23 - 12.00pm (1-5)
7/24 - 1:30 - (6-10)

Date: 7/23/14
Fishpond: Hekia

Initial Observations:

Start Time	8 15am	% Cloud Cover	20%
Finish Time	3:30pm 7/24		
Tide		Precipitation	None / Sunny
Air Temp		Other Comments:	All test done next day tested positive for nitrate. 2??
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	36	49	50	49	35	38	69	36	52	45
Surf. Water Temp °C	24 66	24 42	24 20	24 08	25.07	25 37	24 92	25 66	25 70	25 30
Surf. Salinity	3 81	2 14	0 29	2 55	7 40	13 52	2 90	10 92	10 93	3 95
Surf. DO %	90 2	84 6	54 4	76 1	92 5	100 6	82 4	86 9	90 9	103 0
Surf. DO mg/L	7 30	7 05	5 02	6 18	7 30	7 72	6 69	6 60	6 94	8 60
Surf. pH	7 84	7 61	7 50	7 01	7 71	8 03	7 61	7 87	8 02	8 11
Bottom Temp °C	24 91	25 55	24 85	26 31	25 03	25 46	26 53	25 78	26 31	27 44
Bottom Salinity	4 28	4 78	1 68	11 03	7 32	13 77	13 57	11 21	13 12	12 80
Bottom DO%	79 4	76 3	48 0	69 1	91 4	100 5	69 9	87 8	92 0	85 4
Bottom DO mg/L	6 39	6 01	4 25	5 50	7 24	7 58	5 20	6 72	6 90	6 11
Bottom pH	7 75	7 64	6 90	7 23	7 80	8 09	7 59	7 87	8 03	8 03
NO_3^- (Nitrate) mg/L	0 00	0 00	0 00	0 00	0 40	0 19	0 00	0 26	0 43	0 05
NO_2^- (Nitrate) mg/L	0 003	0 000	0 000	0 000	0 002	0 002	0 000	0 005	0 001	0 002
NH_4 (Ammonia) mg/L	0 11	0 08	0 02	0 05	0 15	0 13	0 11	0 23	0 12	0 10
PO_4^{-3} (Phosphate)	0 18	0 11	0 19	0 03	0 17	0 01	0 20	0 07	0 05	0 11
SiO_2 (Silica) mg/L	19	18	21	18	11	9	14	11	10	15
Turbidity (FAU)	13	20	18	24	9	5	22	10	2	12

Water Quality Sampling Data Sheet:

Date: 7/29/14
Fishpond: He'eia

Keahi
1-3 2/29
4-7 2/30

Initial Observations:

Start Time	1:40	% Cloud Cover 40	
Finish Time			
Tide	↑	Precipitation	Sunny / No rain
Air Temp		Other Comments:	
Wind Speed/Direction			

Site #	1	2	3 (wrong ↑, ← CC)	4	5	6	7	8	9	10
Depth (cm)	40	52	58	57	84	53	47	47	59	63
Surf. Water Temp °C	29.16	14.88	27.00	27.05	27.76	29.52	29.17	28.47	28.41	29.17
Surf. Salinity	19.91	14.62	6.72	14.75	17.70	26.52	21.67	29.42	32.03	23.49
Surf. DO %	140.4	102.4	92.9	100.7	101.5	136.9	127.1	112.1	110.9	151.4
Surf. DO mg/L	9.63	7.49	7.08	7.31	7.07	8.98	10.50	7.31	7.21	10.23
Surf. pH	8.48	8.24	7.78	8.15	8.27	8.60	8.45	8.24	8.36	8.67
Bottom Temp °C	29.33	29.19	29.85	29.35	28.87	29.15	29.40	28.54	28.46	30.01
Bottom Salinity	19.97	21.66	22.20	20.73	22.82	26.53	20.41	29.55	31.83	25.45
Bottom DO%	143.2	153.5	117.7	125.0	130.4	139.9	129.3	113.9	118.9	162.0
Bottom DO mg/L	9.80	10.50	7.68	9.41	8.89	9.25	8.71	7.50	7.66	10.71
Bottom pH	8.51	8.57	8.15	8.50	8.45	8.61	8.40	8.33	8.38	8.67
NO_3^- (Nitrate) mg/L	0.59	0.37	0.16	0.90	0.63	0.32	1.23	0.73	0.89	0.79
NO_2^- (Nitrite) mg/L	0.006	0.004	0.007	0.006	0.009	0.011	0.001	0.003	0.000	0.016
NH_4 (Ammonia) mg/L	0.06	0.00	0.01	0.08	0.03	0.00	0.03	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.13	0.34	0.13	0.18	0.15	0.16	0.48	0.06	0.21	0.05
SiO_2 (Silica) mg/L	7	12	18	15	10	5	10	3	0	5
Turbidity (FAU)	9	6	13	8	4	0	15	0	1	9

Water Quality Sampling Data Sheet:

Date: 7/30/14
Fishpond: Heeia

Initial Observations:

Start Time	7:45a	% Cloud Cover	25% clouds, sunny
Finish Time	8:45a		
Tide	Low & Dropping	Precipitation	No rain past couple days
Air Temp	85°	Other Comments:	
Wind Speed/Direction	5 mph ENE		

Kuli; collected 7/30
Keahi & Kinohi
1-10 08/01

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	42	53	58	49	67	45	40	43	51	50
Surf. Water Temp °C	24.54	23.70	24.21	24.27	24.41	25.72	25.06	26.69	25.80	26.83
Surf. Salinity	19.46	16.27	7.20	17.95	15.31	26.33	19.14	25.73	26.34	26.50
Surf. DO %	88.1	102.8	83.5	96.3	84.5	86.0	96.0	73.4	92.1	81.3
Surf. DO mg/L	6.58	8.36	6.71	7.39	6.45	6.12	7.10	5.11	6.53	5.58
Surf. pH	8.09	8.27	7.66	8.16	8.05	8.30	8.25	8.15	8.37	8.27
Bottom Temp °C	25.44	26.24	26.57	26.72	26.70	25.76	27.35	27.04	26.76	27.06
Bottom Salinity	23.54	7.75	22.00	20.94	24.11	26.85	24.15	26.24	26.78	26.73
Bottom DO%	85.1	64.0	73.1	89.1	44.0	88.9	69.1	70.8	95.8	80.6
Bottom DO mg/L	6.26	4.39	5.55	6.56	4.75	6.26	4.78	4.91	6.54	5.30
Bottom pH	8.12	8.14	7.70	8.23	8.07	8.35	8.15	8.14	8.36	8.27
NO_3^- (Nitrate) mg/L	0.93	0.36	0.00	0.44	0.060	0.81	1.05	0.87	0.77	0.22
NO_2^- (Nitrate) mg/L	0.006	0.007	0.000	0.007	0.002	0.000	0.000	0.010	0.011	0.000
NH_4 (Ammonia) mg/L	0.02	0.02	0.83	0.00	0.01	0.00	0.00	0.00	0.00	0.02
PO_4^{-3} (Phosphate)	0.14	0.16	0.17	0.20	0.03	0.00	0.00	0.51	0.11	0.11
SiO_2 (Silica) mg/L	9	12	17	11	12	3	5	5	3	3
Turbidity (FAU)	10	5	14	8	4	0	4	0	4	0

Water Quality Sampling Data Sheet:

Date: 8/6/14

Fishpond: Heeia

Initial Observations:

Start Time	8:15a	% Cloud Cover	50%
Finish Time	4 :1		
Tide	low low low	Precipitation	1.1 b.f lost right
Air Temp	rising	Other Comments:	
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	35	44	49	49	60	39	37	40	42	49
Surf. Water Temp °C	25.03	24.86	25.21	26.44	25.55	25.49	26.21	25.96	25.83	25.59
Surf. Salinity	19.42	18.35	4.97	22.77	19.85	27.50	22.33	23.98	29.60	23.01
Surf. DO %	97.4	91.2	85.53	84.8	77.9	1065	82.7	94 0	88 1	101.9
Surf. DO mg/L	7.11	6.80	6.01	5.96	6.01	7.70	5.91	6.66	6 10	7.07
Surf. pH	8.08	8.09	7.69	8.07	8.04	8.30	8.09	8 18	8 21	8.31
Bottom Temp °C	25.04	25.17	27.30	26.33	25.97	25.54	26.11	25.97	25.95	25 58
Bottom Salinity	19.55	22.02	21.80	21.91	22.75	27.49	21.21	23.94	29.66	24.75
Bottom DO%	94.9	80.0	46.4	83.1	72.2	108.9	85.3	93.2	85.7	91.9
Bottom DO mg/L	7.01	5.71	3.28	5.91	5.20	7.63	6.04	6.60	5.89	6.62
Bottom pH	8.10	8.14	7.71	8 14	8.04	8.36	8.04	8 19	8 21	8 15
NO_3^- (Nitrate) mg/L	0.58	0.98	0.04	0.59	0.51	0.75	1.08	0.85	0.66	0.93
NO_2^- (Nitrite) mg/L	0.008	0.002	0.000	0.000	0.004	0.000	0.004	0.002	0.002	0.000
NH_4 (Ammonia) mg/L	0.01	0.00	0.04	0.01	0.02	0.00	0.00	0.03	0.00	0.00
PO_4^{-3} (Phosphate)	0.18	0.06	0.33	0.13	1.39	0.17	0.12	0.12	0.0	0.26
SiO_2 (Silica) mg/L	7	11	14	14	12	1	8	3	2	6
Turbidity (FAU)	11	10	48	14	11	3	13	1	0	6

All samples including DI read @ 46 minutes instead of 18.

Water Quality Sampling Data Sheet:

Date 08/11/14
Fishpond He'eia fishpond

08/11
Kinohi collected

08/12
Kinohi & Kelii tested

Initial Observations:

Start Time	01:00 pm	% Cloud Cover	5%
Finish Time			
Tide	rising	Precipitation	none
Air Temp		Other Comments:	clear sky / Hot
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	45	55	72	75	81	54	60	63	80	81
Surf. Water Temp °C	31.83	31.92	32.22	33.85	33.92	32.46	34.17	31.11	28.52	33.31
Surf. Salinity	22.74	26.94	28.43	32.59	33.09	32.75	32.61	33.28	34.14	32.64
Surf. DO %	101.4	108.7	104.3	108.2	132.7	151.2	118.9	120.2	102.9	178.9
Surf. DO mg/L	6.54	7.14	6.05	6.44	7.83	9.27	7.04	7.32	6.62	10.74
Surf. pH	8.09	8.27	8.14	8.14	8.30	8.46	8.15	8.31	8.26	8.52
Bottom Temp °C	33.21	32.80	33.32	33.83	33.02	32.59	34.06	30.68	28.63	32.02
Bottom Salinity	28.29	32.09	32.38	33.01	33.12	32.75	32.48	33.59	34.11	33.41
Bottom DO%	152.6	146.2	152.6	111.8	132.2	100.5	114.7	129.4	116.3	152.8
Bottom DO mg/L	9.06	8.82	8.96	6.91	7.95	7.74	6.70	8.02	7.31	9.51
Bottom pH	8.43	8.41	8.30	8.18	8.29	8.46	8.15	8.34	8.26	8.43
NO_3^- (Nitrate) mg/L	0.90	1.50	1.16	1.82	1.22	1.44	1.10	1.13	1.13	1.12
NO_2^- (Nitrite) mg/L	0.006	0.006	0.002	0.002	0.007	0.002	0.005	0.005	0.004	0.000
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.00	0.03	0.09	0.00	0.00	0.00	0.00	+++	0.04	0.00
SiO_2 (Silica) mg/L	8	4	5	4	3	2	4	2	2	3
Turbidity (FAU)	22	15	31	17	7	2	12	10	0	8

Water Quality Sampling Data Sheet:

Date: 8/13/14

Fishpond: Heeia Kelii collect

Kelii by Kiaohi Testing 08/13/14

Initial Observations:

Start Time	10:56a	% Cloud Cover	30% cloud 70% sun
Finish Time			
Tide	low dropping	Precipitation	Ø
Air Temp	Hot	Other Comments:	
Wind Speed/Direction	5mph ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	41	53	50	55	55	37	43	43	51	48
Surf. Water Temp °C	31.36	30.38	29.90	29.46	30.47	30.87	30.87	31.61	31.51	31.84
Surf. Salinity	33.32	30.06	18.06	24.97	30.82	30.78	32.14	32.85	33.27	33.64
Surf. DO %	120.1	96.2	88.61	95.5	86.7	98.0	120.8	169.0	122.1	118.1
Surf. DO mg/L	7.37	6.55	5.85	6.22	5.49	6.17	7.22	10.24	7.48	7.23
Surf. pH	8.16	8.25	8.02	8.05	8.12	8.16	8.25	8.51	8.27	8.20
Bottom Temp °C	31.43	31.68	30.90	30.64	30.76	30.86	30.94	31.61	31.52	32.02
Bottom Salinity	33.35	31.20	27.74	31.53	31.57	30.78	32.13	32.90	33.19	33.57
Bottom DO%	118.5	131.9	145.0	101.2	74.4	95.5	117.0	166.9	131.5	121.8
Bottom DO mg/L	7.38	8.38	8.30	6.08	4.90	5.96	7.26	10.35	8.11	7.18
Bottom pH	8.18	8.32	7.71	8.16	8.14	8.17	8.26	8.51	8.33	8.23
NO_3^- (Nitrate) mg/L	0.11	0.15	0.52	0.32	0.00	0.00	0.00	0.00	0.00	1.38
NO_2^- (Nitrite) mg/L	0.006	0.010	0.004	0.017	0.008	0.006	0.003	0.000	0.000	0.003
NH_4 (Ammonia) mg/L	0.00	0.00	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.06	0.24	0.18	0.09	0.09	0.25	0.30	0.08	0.07	0.08
SiO_2 (Silica) mg/L	3	4	21	8	4	4	5	3	3	4
Turbidity (FAU)	10	7	38	14	8	9	20	7	5	17

Water Quality Sampling Data Sheet:

Date: 8/19/14
Fishpond: He'eia

Initial Observations:

Start Time	1:30	% Cloud Cover	50%
Finish Time	3:00 pm		
Tide	Rising	Precipitation	No
Air Temp		Other Comments:	
Wind Speed/Direction	E NE		

08/20
Kinohi by Keahi Tested

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	77	88	90	88	105	80	77	79	87	82
Surf. Water Temp °C	27.69	27.16	27.59	27.70	28.08	27.26	28.06	28.23	27.72	27.81
Surf. Salinity *	25.14	19.78	21.36	23.34	28.67	29.85	28.43	33.20	33.65	31.45
Surf. DO %	108.2	105.9	93.8	104.0	98.2	114.5	107.5	121.2	115.4	132.2
Surf. DO mg/L	7.76	7.46	6.41	7.13	6.78	7.71	7.02	7.84	7.53	8.72
Surf. pH	8.32	8.25	7.91	8.23	8.15	8.38	8.26	8.37	8.39	8.47
Bottom Temp °C	27.45	28.04	27.97	28.04	27.75	28.09	28.51	28.29	28.10	27.85
Bottom Salinity	31.88	33.24	29.29	29.66	30.58	33.47	32.69	34.06	34.31	31.48
Bottom DO%	124.9	175.5	101.6	110.3	148.7	143.2	107.8	129.0	130.9	129.1
Bottom DO mg/L	8.25	11.00	6.77	7.33	9.89	9.39	6.98	8.34	8.45	8.51
Bottom pH	8.41	8.58	8.20	8.29	8.48	8.49	8.26	8.41	8.43	8.48
NO_3^- (Nitrate) mg/L *	1.18	0.27	0.08	0.52	0.10	0.07	0.00	0.61	0.38	0.08
NO_2^- (Nitrite) mg/L	0.004	0.000	0.003	0.000	0.002	0.000	0.010	0.000	0.003	0.008
NH_4 (Ammonia) mg/L	---	0.00	---	0.00	0.00	---	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.09	0.00	0.00	0.10	0.00	0.00	0.00	0.03	0.01	0.13
SiO_2 (Silica) mg/L	5	10	14	9	8	3	5	3	3	3
Turbidity (FAU)	12	8	27	10	11	8	5	0	0	0

Water Quality Sampling Data Sheet:

Date: 8/21/14
Fishpond: HE'E IA

Initial Observations:

Start Time	12.43	% Cloud Cover	90%
Finish Time			
Tide	HIGH	Precipitation	Light drizzle
Air Temp		Other Comments:	
Wind Speed/Direction	ENE		

Kinihi & Keahi tested

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	85	95	105	90	118	92	90	91	105	100
Surf. Water Temp °C	26.97	26.91	26.43	26.61	26.64	27.49	26.91	27.16	27.26	27.57
Surf. Salinity	*26.43	*28.10	19.04	19.45	23.22	*34.34	25.03	*33.33	*34.11	*31.45
Surf. DO %	115.2	130.9	91.8	99.1	95.3	130.1	100.1	110.9	117.0	116.5
Surf. DO mg/L	7.86	8.89	6.36	7.15	6.70	8.47	6.93	7.38	7.66	7.73
Surf. pH	8.24	8.42	8.07	8.12	8.24	8.42	8.19	8.36	8.36	8.33
Bottom Temp °C	27.29	27.40	27.28	27.10	27.06	27.54	27.33	27.34	27.51	27.52
Bottom Salinity	32.87	33.82	28.57	28.84	32.55	34.25	33.32	33.44	34.46	33.99
Bottom DO%	131.7	135.2	107.3	97.1	118.7	128.4	92.2	111.9	112.5	135.9
Bottom DO mg/L	8.68	8.65	7.26	6.60	7.87	8.37	6.06	7.35	7.33	8.89
Bottom pH	8.39	8.40	8.22	8.18	8.37	8.43	8.23	8.34	8.37	8.46
NO_3^- (Nitrate) mg/L	0.27	0.63	0.92	0.75	0.75	0.44	1.20	1.39	1.67	1.01
NO_2^- (Nitrate) mg/L	0.00	0.10	0.04	0.00	0.00	0.00	0.00	0.04	0.05	0.04
NH_4 (Ammonia) mg/L	0.09	0.05	0.07	0.10	0.10	0.01	0.12	0.00	0.07	0.00
PO_4^{-3} (Phosphate)	0.00	0.09	0.12	0.21	0.09	0.24	0.00	0.01	0.00	0.00
SiO_2 (Silica) mg/L	5	4	10	10	8	0	7	0	3	5
Turbidity (FAU)	0	3	11	8	9	4	15	0	3	7

* 30

Water Quality Sampling Data Sheet:

Date: 8/25

Fishpond: He'eia

Initial Observations: Kuahi & KINOHI

Start Time	1.18	% Cloud Cover	50%
Finish Time			
Tide	RISING	Precipitation	MORNING SHOWERS
Air Temp		Other Comments:	
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	72	98	98	97	120	81	87	89	95	97
Surf. Water Temp °C	28 95	28 75	28 70	28 88	28 91	29.16	29.16	28.60	28 03	29 42
Surf. Salinity	28 05	24.96	23.95	28 75	29.32	34.21	30.62	32.00	35 18	32.32
Surf. DO %	130.3	116 1	111.5	111.0	121 0	113 3	101.1	113.0	104.1	145.8
Surf. DO mg/L	8 57	8.07	7.55	7.27	8.02	7.27	6 57	7.13	6.71	9.23
Surf. pH	8 38	8.36	8.25	8.23	8.33	8.32	8.22	8.27	8.29	8.56
Bottom Temp °C	29 99	28.98	28.90	29.17	28 83	28.39	28 81	28 51	28.05	28.44
Bottom Salinity	30 .26	32.29	31 71	30.48	32.15	34.12	31.92	33.10	35.19	34 23
Bottom DO%	146 0	130.1	161.6	133 9	144 9	116 4	100.6	116.5	102 7	125 0
Bottom DO mg/L	9.40	8.40	10.48	8.67	9 33	7.48	6.50	7.52	6 62	8.10
Bottom pH	8.47	8.43	10.48	8.26	8.47	8.33	8.23	8.30	8.29	8.38
NO_3^- (Nitrate) mg/L	0.38	0.71.	0.70	0.37	0.00	0.14	0.00	0.67	0.51	0.39
NO_2^- (Nitrite) mg/L	0.045	0.000	0.004	0.007	0.005	0.006	0.006	0.006	0.006	0.006
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.08	0.08	0.12	0.08	0.13	0.09	0.72	0.73	0.17	0.05
SiO_2 (Silica) mg/L	5	7	8	5	4	3	5	5	3	2
Turbidity (FAU)	0	4	8	7	7	0	8	13	0	0

Water Quality Sampling Data Sheet:

Date: 8/28/14

Fishpond: Heeia

Initial Observations: Reli gathered

Start Time	8 15a	% Cloud Cover	50%
Finish Time	9:15a		
Tide	dropping	Precipitation	None today, light yesterday
Air Temp		Other Comments:	
Wind Speed/Direction	0-5 mph ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	52	62	68	58	85	56	50	53	64	57
Surf. Water Temp °C	26.16	25.70	26.29	26.68	26 36	26,37	27,32	27.57	26.66	26.75
Surf. Salinity	27.01	27.03	21.91	25.79	25.32	27.74	31,83	32.53	31.00	29.12
Surf. DO %	79.6	83.4	71.7	66.1	78 6	84,4	58.0	61.3	82.2	77.3
Surf. DO mg/L	5.56	5 82	5.21	4.56	5.40	5.86	3.71	4 07	5.62	5.27
Surf. pH	8.13	8.23	7.96	8.05	8.14	8,23	8.07	8,08	8.27	8.23
Bottom Temp °C	27.12	26.35	28.17	27,60	27.23	26.74	27.74	27.66	27.25	27.71
Bottom Salinity	29.30	28.79	29,80	30.01	32.84	28.81	32.40	32 85	32,81	31.14
Bottom DO%	61.8	74.4	38.7	43.6	37.0	82.1	51.7	57.4	76.7	64.1
Bottom DO mg/L	4.19	4.84	2.61	2.95	2.42	5.57	3.41	3.77	5.12	4.23
Bottom pH	8.08	8.11	7.82	7,97	7.98	8.27	8 08	8.09	8.24	8.20
NO_3^- (Nitrate) mg/L	0.87	0.44	0.46	0.57	0.69	0.00	0.46	0.00	0.72	0.16
NO_2^- (Nitrate) mg/L	0.007	0.001	0.004	0.004	0.002	0.002	0.004	0.012	0.011	0.003
NH_3 (Ammonia) mg/L	0.00	---	0.00	0.14	0.02	---	0.03	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.47	1.46	0.53	1.16	+++	0.99	0.75	0.74	+++	+++
SiO_2 (Silica) mg/L	4	4	14	4	5	5	4	3	3	6
Turbidity (FAU)	3	0	23	3	1	0	8	2	0	1

Water Quality Sampling Data Sheet:

Date: 9/2/14

Fishpond: _____

Initial Observations:

Start Time	1:15	% Cloud Cover	30%
Finish Time			
Tide	High	Precipitation	Morning showers
Air Temp		Other Comments:	
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	77	86	98	88	107	78	71	78	91	81
Surf. Water Temp °C	30.04	29.24	29.11	29.70	29.05	29.67	30.44	29.93	29.55	30.29
Surf. Salinity	24.01	13.62	25.58	29.23	25.40	30.99	31.37	32.65	31.91	28.92
Surf. DO %	86.0	79.4	75.1	101.2	73.3	73.5	61.9	88.5	88.8	82.3
Surf. DO mg/L	5.76	5.55	5.20	4.37	4.90	4.73	4.01	5.73	5.68	5.36
Surf. pH	7.77	7.75	7.76	7.70	7.72	7.77	7.60	7.88	7.86	7.83
Bottom Temp °C	31.42	30.51	30.09	30.11	30.17	30.15	30.36	30.61	30.60	31.16
Bottom Salinity	33.13	34.50	32.17	32.57	33.12	32.15	32.57	34.52	34.36	33.17
Bottom DO%	146.0	137.9	59.2	67.5	115.7	78.0	100.12	141.1	149.9	126.9
Bottom DO mg/L	8.77	8.52	3.76	4.24	7.22	4.95	6.30	8.70	8.99	7.68
Bottom pH	8.11	8.14	7.67	7.79	8.05	7.95	7.90	8.13	8.16	8.19
NO_3^- (Nitrate) mg/L	0.85	1.09	0.32	0.63	0.82	0.69	0.70	1.30	0.48	0.45
NO_2^- (Nitrite) mg/L	0.010	0.016	0.011	0.007	0.000	0.001	0.007	0.003	0.005	0.007
NH_4 (Ammonia) mg/L	0.00	0.00	0.01	0.00	0.05	0.07	0.00	0.04	0.00	0.00
PO_4^{-3} (Phosphate)	0.19	0.66	0.71	0.49	0.22	0.26	0.18	0.51	+++	0.26
SiO_2 (Silica) mg/L	10	10	12	4	4	3	2	3	3	6
Turbidity (FAU)	0	0	13	24	0	0	0	18	0	0

Nitrate start: 1.30pm
Turbidity start: 205pm.
Nitrite start: 1.50pm test 210pm.
Silica test 2.37pm.

Ammonia start: 2.50pm test start: 3.08pm

Start: 1.30pm
Pau: 3.12 pm

Water Quality Sampling Data Sheet:

Date: 9/8/14

Fishpond: _____

9/8 - collected H_2O
9/9 - testing

Initial Observations:

Start Time	1:30pm	% Cloud Cover	90%
Finish Time	01:30 pm 09/09		
Tide	HIGH	Precipitation	None
Air Temp		Other Comments:	High Thin layer of Clouds across sky "so many"
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	99	107	102	108	147	101	100	104	112	113
Surf. Water Temp "C	28.13	28.39	29.85	29.64	29.43	34.85	29.87	29.20	28.44	29.35
Surf. Salinity	25.11	22.11	26.86	30.33	29.42	34.86	31.57	33.42	34.65	30.84
Surf. DO %	106.2	95.8	82.9	107.4	101.8	103.9	101.6	103.1	105.6	123.6
Surf. DO mg/L	7.23	6.61	6.02	6.71	7.06	6.62	6.49	6.57	6.68	8.14
Surf. pH	7.93	7.84	7.87	7.92	8.04	7.97	7.94	7.95	8.01	8.10
Bottom Temp "C	29.41	28.74	29.08	29.46	29.25	28.85	29.69	29.14	28.95	29.11
Bottom Salinity	32.69	34.34	32.66	32.27	32.61	34.83	32.62	34.30	34.05	34.37
Bottom DO%	147.5	103.1	103.8	113.6	123.7	101.1	113.5	106.6	103.2	141.8
Bottom DO mg/L	9.37	6.59	6.99	7.10	7.84	6.43	7.24	6.73	6.58	8.96
Bottom pH	8.16	7.99	8.00	7.99	8.08	8.00	7.98	8.00	8.02	8.13
NO_3^- (Nitrate) mg/L	1.02	0.64	0.45	0.34	0.00	0.00	0.00	0.48	0.11	0.00
NO_2^- (Nitrite) mg/L	0.006	0.004	0.004	0.008	0.000	0.006	0.006	0.004	0.005	0.003
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00
PO_4^{-3} (Phosphate)	0.11	0.08	0.24	0.13	0.30	0.21	0.07	0.02	0.22	0.08
SiO_2 (Silica) mg/L	8	11	17	5	8	2	4	3	2	5
Turbidity (FAU)	2	8	18	2	1	7	3	16	6	12

Water Quality Sampling Data Sheet:

Date: 9/11/14
Fishpond: Heeia

Tested 9/11 – 12-1:30

Initial Observations:

Start Time	8:20am 9/11	% Cloud Cover	20 % + sunny as hell
Finish Time	01:30p 9/11		
Tide	Dropping	Precipitation	None
Air Temp	Hot	Other Comments:	
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	59	66	70	63	90	60	67	58	78	64
Surf. Water Temp °C	26.63	26.45	26.36	26.47	27.60	27.53	27.70	27.89	27.57	27.88
Surf. Salinity	27.35	23.86	13.01	24.35	22.50	30.76	29.01	32.66	32.67	32.60
Surf. DO %	80.5	90.1	61.2	81.7	75.6	78.1	72.0	76.0	78.2	70.4
Surf. DO mg/L	5.32	6.34	4.61	5.56	5.32	5.18	4.85	4.92	5.14	4.61
Surf. pH	7.66	7.79	7.13	7.70	7.20	7.85	7.77	7.84	7.87	7.85
Bottom Temp °C	28.71	28.21	28.55	28.86	28.37	28.03	28.56	28.50	28.17	28.63
Bottom Salinity	34.33	30.99	32.82	33.18	33.97	31.81	32.49	33.78	33.45	34.38
Bottom DO%	65.1	54.0	31.0	46.5	36.1	70.7	51.0	70.2	75.0	69.9
Bottom DO mg/L	4.14	3.62	1.87	3.07	2.30	4.55	3.28	4.55	4.92	4.50
Bottom pH	7.79	7.70	7.53	7.68	7.69	7.87	7.74	7.83	7.89	7.88
NO_3^- (Nitrate) mg/L	1.55	0.77	0.63	1.08	0.86	0.85	0.98	1.24	1.37	1.20
NO_2^- (Nitrite) mg/L	0.005	0.002	0.000	0.000	0.000	0.004	0.000	0.008	0.008	0.009
NH_4 (Ammonia) mg/L	0.00	0.00	---	0.00	0.00	0.00	0.02	0.00	---	0.00
PO_4^{-3} (Phosphate)	0.69	0.40	0.46	1.26	0.56	1.01	+++	0.28	0.89	2.27
SiO_2 (Silica) mg/L	9	10	17	11	11	6	8	4	5	7
Turbidity (FAU)	1	23	15	0	7	6	13	3	5	2

Water Quality Sampling Data Sheet:

Date: 9/16/14
Fishpond: Helela

Initial Observations:

Start Time	7:15am	% Cloud Cover	50%
Finish Time	4:05pm		
Tide	Rising	Precipitation	None
Air Temp		Other Comments:	Rain early morning
Wind Speed/Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
		83								
Depth (cm)	66	83	79	89	113	74	74	77	84	87
Surf. Water Temp °C	27.95	28.42	29.31	28.83	27.74	28.68	28.86	28.46	29.26	28.22
Surf. Salinity	27.51	31.38	3150	30.58	28.16	33.35	30.65	31.08	34.34	29.77
Surf. DO %	46.7	41.9	38.2	52.2	65.0	60.4	54.6	62.9	82.7	52.6
Surf. DO mg/L	3.18	2.67	2.47	3.41	4.42	3.89	3.70	4.09	5.33	3.46
Surf. pH	7.51	7.64	7.61	7.70	7.69	7.79	7.69	7.75	7.90	7.69
Bottom Temp °C	28.53	28.34	29.67	29.77	28.67	28.91	29.53	29.06	29.47	29.30
Bottom Salinity	31.38	32.93	32.23	32.34	32.11	34.56	33.54	34.06	34.99	34.63
Bottom DO%	38.3	39.0	30.5	32.0	37.7	58.6	31.9	61.2	79.3	74.1
Bottom DO mg/L	2.51	2.71	2.00	2.84	2.49	3.77	2.04	3.89	5.00	4.66
Bottom pH	7.61	7.66	7.58	7.60	7.63	7.79	7.61	7.80	7.92	7.85
NO_3^- (Nitrate) mg/L	0.09	0.16	0.00	0.75	0.22	0.23	0.87	0.72	0.99	0.80
NO_2^- (Nitrite) mg/L	0.005	0.004	0.004	0.007	0.004	0.005	0.007	0.000	0.000	0.003
NH_4 (Ammonia) mg/L	0.02	0.00	0.00	0.00	0.01	0.00	0.05	0.23	0.18	0.05
PO_4^{-3} (Phosphate)	0.15	0.50	0.00	0.07	0.17	0.34	0.44	0.16	0.21	0.08
SiO_2 (Silica) mg/L	8	6	7	6	5	4	5	2	1	5
Turbidity (FAU)	3	9	15	9	9	0	12	1	4	8

Water Quality Sampling Data Sheet:

Date: 9/17/14
Fishpond: _____

Initial Observations:

Start Time	7.10	% Cloud Cover	30%
Finish Time			
Tide	Lower Rising	Precipitation	Early morning
Air Temp		Other Comments:	
Wind Speed/Direction	E / still		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	50	60	72	61	94	57	54	58	74	66
Surf. Water Temp °C	27.65	29.48	29.85	29.54	28.64	27.43	27.47	28.54	29.30	27.52
Surf. Salinity	29.20	31.60	31.08	30.08	29.90	30.05	26.71	32.43	34.77	29.54
Surf. DO %	65.7	33.0	47.1	42.0	59.0	67.7	70.0	72.6	82.5	72.1
Surf. DO mg/L	4.40	2.13	3.05	3.03	3.86	4.50	4.85	4.69	5.21	4.83
Surf. pH	7.73	7.61	7.68	7.61	7.69	7.78	7.70	7.83	7.91	7.76
Bottom Temp °C	28.20	29.55	30.29	29.88	30.73	27.86	28.92	28.94	29.31	28.02
Bottom Salinity	29.75	31.67	31.66	30.88	32.01	30.90	31.75	33.21	34.82	29.88
Bottom DO%	57.0	25.4	33.5	42.0	32.1	49.4	56.7	72.0	82.0	52.0
Bottom DO mg/L	3.77	1.63	2.12	2.82	2.06	3.26	3.67	4.66	5.23	3.44
Bottom pH	7.63	7.60	7.58	7.58	7.62	7.70	7.70	7.84	7.92	7.73
NO_3^- (Nitrate) mg/L	0.47	1.02	0.58	1.12	0.27	1.01	0.91	0.97	1.27	0.75
NO_2^- (Nitrite) mg/L	0.006	0.006	0.002	0.000	0.000	0.003	0.005	0.006	0.005	0.003
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	2.20	0.16	0.31	0.19	1.51	+++	0.44	0.61	0.49	0.22
SiO_2 (Silica) mg/L	~~7~~	2	5	6	4	2	4	3	2	4
Turbidity (FAU)	5	0	9	0	0	0	0	0	0	0

Turbidity (FAU) 5 0 6 3 6 4 5 0 0 1

→ change DI water
→ recalibrate, turn over before testing. Lone time, inverted the vi
→ re-test

Water Quality Sampling Data Sheet:

Date: 9/22
Fishpond: Helen

Tested
9/23 - 7:00 - 9:00
Keahi Keliʻi

Initial Observations:

Start Time	1:15 pm - 2:30	% Cloud Cover	30%
Finish Time			
Tide	High @ 1.42pm 1.99'	Precipitation	None
Air Temp		Other Comments:	
Wind Speed/Direction	ENE 8-13 kts		

30.34

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	87	99	100	98	117	90	88	89	102	99
Surf. Water Temp °C	32.78	30.39	29.01	30.15	30.31	[illegible]	30.33	30.10	30.30	31 30
Surf. Salinity	32.65	22.45	1927	30.92	32.48	34.59	31.76	34.09	34.92	34.411
Surf. DO %	117.0	113.4	93.6	108.3	101.4	119.1	112.0	100.8	126.1	132.5
Surf. DO mg/L	7.06	8.45	6.52	6.73	6.41	7.42	7.07	6.67	7.82	8.13
Surf. pH	7.86	7.90	7.95	7.98	7.89	BDO	7.93	7.95	8 07	8.07
Bottom Temp °C	31.38	30.64	30.59	31.49	30.23	30.41	30.51	30.12	30.39	31 28
Bottom Salinity	33.50	31.39	33.32	32.85	33.35	34.61	33.06	34.11	34 98	34.40
Bottom DO%	162.1	148.8	118.8	129.9	102.8	118.6	118.9	104 9	127.1	130 9
Bottom DO mg/L	10.02	9.21	7.46	7.57	6.43	7.36	7 40	6.68	7.85	8.22
Bottom pH	8.04	8.04	7.95	7.99	7.92	8 01	7.97	7.95	8.07	8.06
NO_3^- (Nitrate) mg/L	0.70	0-82	0.49	0.76	0.77	1.29	1.00	0.45	0.28	1.16
NO_2^- (Nitrite) mg/L	0.001	0.004	0.003	0.006	0.004	0.004	0.000	0.000	0.002	0.002
NH_4 (Ammonia) mg/L	0.00	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.02	0.06	0.15	0.04	0.18	0.10	0.10	0.12	0.14	0.14
SiO_2 (Silica) mg/L	4	9	13	7	8	3	5	3	2	3
Turbidity (FAU)	15	6	13	4	6	0	1	0	0	0

Water Quality Sampling Data Sheet:

Date: 9/25/14

Fishpond: Heeia

Initial Observations:

Start Time	7.40a	% Cloud Cover	50% clouds
Finish Time	9.00a	collecting time	
Tide	Dropping	Precipitation	0
Air Temp		Other Comments:	
Wind Speed/Direction			

Tested 9/26
Start - 12pm
End - 1:45p

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	52	63	61	61	80	57	49	53	57	58
Surf. Water Temp °C	26.38	26.16	26.93	26.81	26.80	27.71	27.77	28.11	27.66	27.48
Surf. Salinity	25.30	22.40	20.93	24.76	26.40	31.97	29.71	31.45	32.70	31.47
Surf. DO %	96.0	88.2	83 8	91.2	92.3	76.8	73.1	61.6	86.6	83.1
Surf. DO mg/L	6.78	6.30	5.67	6.33	6.37	4.89	4.88	4.03	5.68	5.49
Surf. pH	7,76	7.67	7.41	7.80	7.83	7.78	7.76	7.72	7.85	7.85
Bottom Temp °C	26.85	27.42	27.89	27.50	28.55	27.74	32.32	28.23	27.70	27.47
Bottom Salinity	25.41	26.33	30.53	29.64	32.55	32.02	40.0	31.56	32.72	31.37
Bottom DO%	85.5	90.0	73.2	70.8	32.7	73 8	2.55	58.9	86.6	77.5
Bottom DO mg/L	5.75	6.08	4.82	4.67	2.09	4.85	4.47	3 85	5.74	5.59
Bottom pH	7.80	7.76	7.75	7.80	7.66	7.79	7.73	7.72	7.85	7.86
NO_3^- (Nitrate) mg/L	0.80	0.78	0.55	0.86	0.80	0.51	0.66	0.22	1.47	0.73
NO_2^- (Nitrite) mg/L	0.002	0.006	0.003	0.000	0.001	0.000	0.000	0.025	0.005	0.000
NH_4 (Ammonia) mg/L	0.00	0.05	0.03	0.00	0.01	0.01	0.02	0.02	0.02	0.00
PO_4^{-3} (Phosphate)	0.11	1.57	+++	0.20	+++	0.20	0.19	0.55	0.00	0.19
SiO_2 (Silica) mg/L	7	6	4	7	6	2	4	6	2	4
Turbidity (FAU)	0	11	42	24	3	10	4	2	0	0

Water Quality Sampling Data Sheet:

Date: 09/30/14
Fishpond: He'eIA

Tested - Keahi
Wed. 9/31
1:00-3:30

Initial Observations:

Start Time	1:00pm - 2:20	% Cloud Cover	80%
Finish Time			
Tide	1.3 ft.	Precipitation	RAINING
Air Temp	32°C	Other Comments:	Thunder, RAIN coming from mauka 2:00 - blazing Hot
Wind Speed/Direction	ENE 2kts		

7 29 12

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	66	72	83	75	98	68	62	65	70	75
Surf. Water Temp °C	29.61	29.56	29.63	29.36	29.65	29.13	29.36	28.81	28.75	29.69
Surf. Salinity	31.50	31.94	32.08	30.24	32.64	30.25	27.29	27.01	29.08	31.40
Surf. DO %	75.3	76.3	60.3	73.7	62.0	86.0	75.3	89.7	93.3	86.0
Surf. DO mg/L	4.93	4.85	3.83	4.77	3.95	5.74	4.90	5.96	6.13	5.57
Surf. pH	7.76	7.79	7.74	7.74	7.80	7.89	7.72	7.80	7.89	7.86
Bottom Temp °C	30.25	30.06	29.85	30.07	29.46	29.69	29.97	29.63	28.88	30.30
Bottom Salinity	34.01	34.00	37.57	33.25	34.06	33.14	32.83	33.20	29.67	33.62
Bottom DO%	106.1	107.2	47.4	71.6	40.5	100.8	61.7	84.3	88.1	124.2
Bottom DO mg/L	6.58	6.74	2.98	4.51	2.56	6.37	3.90	5.35	5.76	7.58
Bottom pH	7.95	8.00	7.67	7.78	7.76	7.95	7.74	7.89	7.89	8.03
NO_3^- (Nitrate) mg/L	0.46	1.14	0.85	0.67	1.27	0.84	0.82	0.90	0.76	1.08
NO_2^- (Nitrite) mg/L	0.001	0.003	0.008	0.013	0.008	0.002	0.000	0.006	0.001	0.002
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.02	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.11	0.14	0.27	0.25	0.17	0.24	0.21	0.13	0.18	0.23
SiO_2 (Silica) mg/L	4	3	5	8	5	4	6	7	6	5
Turbidity (FAU)	5	4	4	9	6	7	8	4	7	4

Water Quality Sampling Data Sheet:

Date: 10/2/11
Fishpond: Hokuula
Initial Observations: Kelii

Keahi 1/3 kinohi tetid
Th. 1013. 100-3:30 PM

Start Time	8:37 am	% Cloud Cover	10 %
Finish Time			
Tide	High	Precipitation	
Air Temp		Other Comments:	
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	80	89	98	91	131	85	85	88	95	96
Surf. Water Temp °C	29.18	27.72	29.68	27.08	28.24	29.55	29.76	28.26	28.95	29.61
Surf. Salinity	27.54	21.63	13.16	14.03	18.89	25.83	27.21	28.60	34.43	30.34
Surf. DO %	68.9	92.6	64.3	92.3	79.1	83.8	68.7	86.5	91.7	69.5
Surf. DO mg/L	4.61	6.44	4.57	6.80	5.59	5.40	4.49	5.79	5.83	4.50
Surf. pH	7.76	7.86	7.28	7.67	7.71	7.80	7.74	7.87	7.91	7.82
Bottom Temp °C	30.43	29.80	30.24	30.37	29.93	29.00	30.47	28.76	29.00	28.84
Bottom Salinity	31.76	32.04	30.27	30.79	31.57	33.94	31 00	31.88	34.76	30.78
Bottom DO%	45.4	47.5	47.1	30.4	46.2	101.4	33.0	88.5	87.1	90.1
Bottom DO mg/L	2.87	3.34	3.00	1.89	2.93	6.48	2.06	5.76	554	5.86
Bottom pH	7.71	7.80	7.65	7.56	7.65	7.96	7.55	7.91	7.91	7.98
NO_3^- (Nitrate) mg/L	0.71	1.12	0.63	0.52	0.51	0.07	0.41	0.68	1.31	0.67
NO_2^- (Nitrite) mg/L	0.004	0.003	0.008	0.004	0.006	0.001	0.001	0.000	0.002	0.007
NH_4 (Ammonia) mg/L	0.00	0.00	0.03	0.01	0.00	0.01	0.00	0.00	0.03	0 00
PO_4^{-3} (Phosphate)	1.32	0.20	2.24	0.38	0.86	0.13	0.61	0.95	0.66	1.78
SiO_2 (Silica) mg/L	12	7	15	13	12	9	8	7	3	6
Turbidity (FAU)	0	10	10	9	5	0	1	0	0	0

Water Quality Sampling Data Sheet:

Date: 10/7/14

Fishpond: Hale'ia

10/8 - tested 1:00-3:00 PM

Initial Observations:

Start Time	1 30 - 3:00	% Cloud Cover	50%
Finish Time	3:00 10/8		
Tide	Flah	Precipitation	9:00-12:00 Light Rain
Air Temp	29°C	Other Comments:	Vog
Wind Speed/Direction	2 Kts		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	88	96.	101	96	128	92	89	88	103	92
Surf. Water Temp °C	29.94	30.10	30.35	30.29	30.05	29.23	30.76	28.97	28.80	29.00
Surf. Salinity	33.80	34 19	33 81	33.69	33.42	34.44	33.60	32 59	32.53	33.54
Surf. DO %	97.1	100.5	111.3	96.1	113.2	105.0	85.4	103.5	111.8	108.5
Surf. DO mg/L	6.15	6.43	6.96	6.01	7.12	6.65	6.30	6.63	7.22	6.94
Surf. pH	7.96	7.98	8.03	7.93	8 02	8.01	7.85	8.00	8.04	8.05
Bottom Temp °C	30.30	29.69	30.50	30.52	29.92	29.23	30.56	29.19	29.04	29.42
Bottom Salinity	34 22	34.59	34.08	33.84	34 12	34.54	33.83	33.94	34 67	34.06
Bottom DO%	76.1	103.1	47.7	115.7	113.6	97.5	89.7	100.6	108.6	103.3
Bottom DO mg/L	4 71	6.41	6.08	7.21	7 13	6 18	4.33	6.39	6 77	6.53
Bottom pH	7.92	8 01	7.87	7.99	8 04	8.00	7.85	7.98	8.03	8.05
NO_3^- (Nitrate) mg/L	0.43	0.416	0 74	1 12	0.82	0.80	0.74	0.68	0 59	1 05
NO_2^- (Nitrite) mg/L	0 013	0 000	0.006	0.000	0.007	0 005	0.000	0 007	0.009	0.004
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0 00
PO_4^{-3} (Phosphate)	0.09	0.25	0.10	0.30	0.10	0.14	0.17	0.15	0.12	0.19
SiO_2 (Silica) mg/L	2	3	3	2	2	2	2	3	4	1
Turbidity (FAU)	4	2	3	10	0	0	3	0	0	0

Water Quality Sampling Data Sheet:

Date: 10/09/14
Fishpond: Heeia

Testing Kinahi & Koahi
10/10/14
1:30pm - 03:15pm

Initial Observations:

Start Time	1:15 pm	% Cloud Cover	70%
Finish Time	3:15 pm		
Tide	0.5m 2:34 pm	Precipitation	None
Air Temp	28°C	Other Comments:	
Wind Speed/Direction	13 kts		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	70	82	93	86	115	78	79	77	89	86
Surf. Water Temp °C	28.15	29.08	29.97	30.40	30.40	29.75	30.77	30.72	29.43	29.91
Surf. Salinity	26.15	28.95	23.01	30.93	31.63	34.16	31.56	33.66	34.67	34.00
Surf. DO %	107.3	132.4	100.9	119.0	132.8	127.8	125.1	138.0	111.4	148.9
Surf. DO mg/L	7.23	8.35	6.62	7.54	8.36	8.05	7.87	8.57	7.05	9.41
Surf. pH	7.85	8.04	7.92	7.97	8.03	8.02	7.97	8.04	7.99	8.14
Bottom Temp °C	29.82	29.78	30.41	30.49	30.69	29.79	30.78	30.72	29.45	29.92
Bottom Salinity	32.29	33.05	32.56	31.70	32.88	34.16	31.61	33.67	34.67	33.99
Bottom DO%	175.6	155.3	139.1	130.9	151.2	128.9	127.8	139.4	116.5	146.9
Bottom DO mg/L	11.15	9.85	8.72	8.24	9.42	8.10	7.99	8.66	7.32	9.22
Bottom pH	8.18	8.18	8.04	8.02	8.08	8.03	7.97	8.04	7.98	8.15
NO_3^- (Nitrate) mg/L	0.92	0.34	0.93	0.88	0.81	0.51	0.76	0.82	1.31	0.84
NO_2^- (Nitrite) mg/L	0.000	0.000	0.004	0.006	0.002	0.004	0.002	0.005	0.003	0.013
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	2.26	0.55	1.33	1.60	0.86	1.07	1.97	1.07	0.39	+++
SiO_2 (Silica) mg/L	8	10	10	5	6	5	5	4	3	4
Turbidity (FAU)	8	6	11	1	6	0	5	0	0	7

Water Quality Sampling Data Sheet:

Date: 10/13/14
Fishpond: Heeia

Initial Observations:

Start Time	9:45a	% Cloud Cover	60-70%
Finish Time	03:40pm		
Tide	dropping	Precipitation	
Air Temp		Other Comments:	
Wind Speed/Direction	NE consistent 10 mph		

Testing Finahi by Keahi
10/14/14
01:00pm - 02:45pm

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	70	81	99	80	93	75	66	69	74	74
Surf. Water Temp °C	25.19	25.31	25.71	25.87	25.86	26.07	26.06	27.03	26.20	25.99
Surf. Salinity	23.43	22.A	20.15	26.60	27.91	27.75	28.09	32.92	30.39	27.63
Surf. DO %	90.7	110.4	93.4	90.9	95.0	895	84.6	85.9	95.8	96.9
Surf. DO mg/L	6.55	7.98	6.56	6.36	6.61	6.56	5.86	5.69	6.45	6.71
Surf. pH	7.91	7.96	7.72	7.81	786	7.87	7.82	7.83	7.92	7.94
Bottom Temp °C	25.85	26.13	26.28	26.17	26.19	26.73	26.71	27.14	26.21	27.23
Bottom Salinity	28.92	30.59	29.68	29.41	30.15	31.65	31.28	33.10	30.44	32.41
Bottom DO%	75.9	85.1	80.5	80.2	94.8	88.1	73.4	88.9	92.5	95.0
Bottom DO mg/L	5.25	5.75	5.61	5.17	6.47	5.88	5.01	5.86	6.30	6.31
Bottom pH	7.89	7.86	7.77	7.78	7.87	7.89	7.80	7.86	7.91	7.94
NO_3^- (Nitrate) mg/L	0.96	0.87	0.68	1.08	0.44	0.68	0.13	0.66	1.33	0.51
NO_2^- (Nitrite) mg/L	0.006	0.000	0.005	0.003	0.001	0.000	0.006	0.008	0.003	0.002
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	2.18	+++	0.25	1.21	+++	+++	+++	0.59	0.28	2.09
SiO_2 (Silica) mg/L	9	8	13	8	7	10	6	5	5	7
Turbidity (FAU)	13	10	8	2	8	0	6	7	0	4

Water Quality Sampling Data Sheet:

Date: 10/15/14
Fishpond: He'eIA
Initial Observations:

Start Time	1:09 pm	Cloud Cover	40%
Finish Time	3.00 10/16		
Tide	1.3 ft	Precipitation	None
Air Temp		Other Comments:	Very Hot Light winds
Wind Speed/Direction	ENE ??		

10/16 tested Kirohi & Keahi

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	68	77	74	74	104	68	54	61	83	71
Surf. Water Temp °C	28.39	28.30	27.94	28.73	28.12	28.46	28.58	28.71	29.04	29.25
Surf. Salinity	22.08	25.15	30.39	31.15	27.61	24.90	29.86	30.41	26.94	24.28
Surf. DO %	105.3	95.5	64.0	86.0	63.3	84.4	53.4	62.5	87.1	99.7
Surf. DO mg/L	7.28	6.46	4.27	5.37	4.22	5.71	3.35	4.07	5.72	6.71
Surf. pH	7.79	7.92	7.72	7.78	7.85	7.82	7.69	7.73	7.85	7.94
Bottom Temp °C	29.63	28.67	28.61	28.68	28.53	28.65	29.91	29.81	29.25	30.25
Bottom Salinity	31.03	33.22	32.84	32.97	33.33	30.01	32.59	33.58	31.07	32.89
Bottom DO%	121.9	120.6	116.3	97.6	110.4	72.9	90.1	153.7	72.2	196.6
Bottom DO mg/L	7.79	7.76	7.55	6.28	7.16	4.72	5.74	9.69	4.68	12.43
Bottom pH	8.06	8.02	7.92	7.84	7.96	7.94	7.82	8.07	7.80	8.24
NO_3^- (Nitrate) mg/L	0.80	0.81	0.00	0.47	0.26	1.03	0.48	0.04	1.07	0.84
NO_2^- (Nitrite) mg/L	0.001	0.001	0.000	0.004	0.006	0.001	0.000	0.002	0.000	0.003
NH_4 (Ammonia) mg/L	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	1.63	+++	1.21	+++	1.52	+++	+++	+++	1.25	0.81
SiO_2 (Silica) mg/L	11	8	15	9	9	9	6	5	6	0
Turbidity (FAU)	4	8	7	5	2	0	6	4	1	7

Water Quality Sampling Data Sheet:

Date: 10/20/14
Fishpond: He'e la
Initial Observations: Day after STORM

Start Time	1:30 10/20	% Cloud Cover	80%
Finish Time	2:30 10/21		
Tide		Precipitation	None
Air Temp		Other Comments:	Rain all day Sat.-Sun
Wind Speed/Direction			

10/21
Kealy & Kinohi
Tested
01:00pm - 02:30pm

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	74	83	83	70	102	74	68	70	77	75
Surf. Water Temp °C	28.01	27.03	27.50	27.84	27.68	27.56	27.80	27.93	27.23	27.91
Surf. Salinity	6.67	2.36	3.09	6.36	5.37	6.18	6.03	7.53	18.71	4.23
Surf. DO %	93.8	75.1	76.0	95.2	88.6	94.9	92.0	96.6	92.4	87.1
Surf. DO mg/L	7.07	5.91	5.61	7.24	6.80	7.23	6.99	7.63	6.66	6.70
Surf. pH	7.63	7.47	7.34	7.63	7.61	7.81	7.60	7.71	7.82	7.66
Bottom Temp °C	27.82	26.73	27.42	27.21	27.52	27.23	27.64	27.44	27.35	27.29
Bottom Salinity	23.70	25.41	27.87	22.48	28.40	26.65	24.78	27.40	26.79	26.43
Bottom DO%	66.3	107.5	81.9	90.8	113.4	136.6	94.41	137.7	138.9	155.3
Bottom DO mg/L	4.62	7.44	5.54	6.37	7.67	9.34	6.48	9.36	9.57	10.65
Bottom pH	7.81	7.93	7.47	7.77	7.87	8.01	7.82	8.08	8.15	8.21
NO_3^- (Nitrate) mg/L	0.00	0.00	0.00	0.00	0.34	0.00	0.00	0.26	0.20	0.10
NO_2^- (Nitrite) mg/L	0.000	0.004	0.000	0.000	0.002	0.003	0.000	0.007	0.012	0.000
NH_4 (Ammonia) mg/L	0.04	0.05	0.05	0.05	0.05	0.05	0.04	0.04	0.04	0.05
PO_4^{-3} (Phosphate)	2.26	1.04	1.28	0.47	1.40	+++	0.95	0.54	1.56	0.93
SiO_2 (Silica) mg/L	17	20	20	17	18	17	16	15	14	18
Turbidity (FAU)	15	17	17	15	16	10	20	17	12	33

Water Quality Sampling Data Sheet:

Date: 10/22/14
Fishpond: He'e19

Tested 10/23
2:00 - 3:30
Kealii Kinoshita

Initial Observations:

Start Time	12:45 - 2:00	% Cloud Cover	85%
Finish Time	3:30 1423		
Tide	1:14pm | 6.8ft	Precipitation	Light Drizzle
Air Temp	28°C	Other Comments:	High tide,
Wind Speed/Direction	ENE 10kts		No moon

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	72	82	90	96	111	79	74	76	90	84
Surf. Water Temp °C	28.74	28.40	27.99	27.80	27.95	27.96	28.59	28.56	28.11	28.96
Surf. Salinity	24.65	26.93	10.96	19.50	11.11	12.60	16.81	28.99	31.14	26.42
Surf. DO %	84.4	95.0	95.4	95.0	93.8	91.6	88.9	112.2	114.3	99.2
Surf. DO mg/L	5.41	6.54	7.03	6.98	6.90	6.70	6.25	7.39	7.50	6.62
Surf. pH	7.88	7.96	7.82	7.96	7.87	7.94	7.90	8.00	8.04	8.01
Bottom Temp °C	29.36	28.97	28.72	28.91	28.66	28.47	29.04	28.65	29.26	29.11
Bottom Salinity	30.59	30.83	28.73	27.53	30.66	32.26	29.55	30.87	32.26	31.27
Bottom DO%	129.9	134.4	97.6	94.2	111.4	123.8	113.9	131.5	121.1	140.9
Bottom DO mg/L	8.43	8.79	6.40	6.66	7.28	7.85	7.45	8.50	7.90	9.23
Bottom pH	8.13	8.14	7.81	7.97	8.00	8.08	7.95	8.09	8.06	8.20
NO_3^- (Nitrate) mg/L	0.26	0.53	0.00	0.41	0.32	0.07	0.13	0.64	0.66	0.79
NO_2^- (Nitrite) mg/L	0.002	0.007	0.004	0.004	0.009	0.004	0.004	0.007	0.004	0.006
NH_4 (Ammonia) mg/L	0.02	0.02	0.02	0.03	0.02	0.01	0.03	0.03	0.00	0.00
PO_4^{-3} (Phosphate)	+++	+++	0.80	+++	0.28	+++	0.81	1.31	0.86	+++
SiO_2 (Silica) mg/L	10	14	13	15	14	14	12	6	3	7
Turbidity (FAU)	2	0	10	2	2	0	0	0	0	0

Water Quality Sampling Data Sheet:

Date: 10/27/14
Fishpond: Heʻeia

tested 10/28
1:00-3:00
Keahi

Initial Observations: Keahi & Girdhi

Start Time	12:30-2:00	% Cloud Cover	75%
Finish Time			
Tide	11.44 1.00ft	Precipitation	Drizzle
Air Temp	28°C	Other Comments:	
Wind Speed/Direction	11kts ENE 2pm		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	52	66	67	64	100	57	53	57	73	66
Surf. Water Temp °C	26.07	25.97	25.30	25.88	26.35	26.91	26.65	27.96	27.70	27.17
Surf. Salinity	17.75	16.70	7.25	15.01	20.52	25.90	21.71	32.58	31.85	26.72
Surf. DO %	90.5	94.2	74.4	98.5	97.1	103.2	95.8	96.3	121.7	111.6
Surf. DO mg/L	6.66	7.35	5.93	7.35	7.02	7.10	6.88	6.28	8.01	7.63
Surf. pH	7.82	7.90	7.39	7.82	7.89	7.95	7.91	7.93	8.09	8.08
Bottom Temp °C	27.60	27.18	28.09	27.11	28.18	28.38	27.52	27.98	27.93	28.56
Bottom Salinity	28.53	25.39	29.26	25.38	33.27	30.86	28.97	32.59	32.21	30.08
Bottom DO%	116.6	95.3	79.1	89.4	79.7	159.9	87.2	94.6	128.0	118.8
Bottom DO mg/L	7.86	6.43	5.26	6.21	5.14	9.87	5.97	6.18	8.38	11.73
Bottom pH	8.07	8.01	7.60	7.88	7.90	8.22	7.92	7.94	8.13	8.33
NO_3^- (Nitrate) mg/L	0.60	0.12	0.33	0.79	0.91	1.21	0.88	1.76	1.25	1.03
NO_2^- (Nitrite) mg/L	0.000	0.004	0.002	0.010	0.003	0.005	0.014	0.003	0.004	0.001
NH_4 (Ammonia) mg/L	0.03	0.03	0.02	0.01	0.03	0.00	0.00	0.00	0.00	0.04
PO_4^{-3} (Phosphate)	2.00	+++	+++	+++	2.05	0.92	0.56	0.81	0.55	0.24
SiO_2 (Silica) mg/L	13	15	20	13	14	6	10	5	3	5
Turbidity (FAU)	0	3	7	6	8	0	1	4	0	0

Water Quality Sampling Data Sheet:

Date: 10/29/14
Fishpond: Heleia

10/31 tested Keahi & Kinoha 7:00 - 8:30 am

Initial Observations:

Start Time	1:00 pm - 2:30 pm	% Cloud Cover	52%
Finish Time	8:30 am 10/31		
Tide	0.3m	Precipitation	None
Air Temp	28°C	Other Comments:	
Wind Speed/Direction	13 Kts		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	55	70	70	61	80	60	55	56	64	63
Surf. Water Temp °C	27.03	26.57	27.01	27.09	27.27	27.46	27.60	27.82	28.00	27.89
Surf. Salinity	17.37	19.49	20.01	22.15	25.34	26.66	27.86	27.54	28.24	23.80
Surf. DO %	108.5	97.7	92.2	95.3	101.8	102.5	125.0	117.4	128.4	112.7
Surf. DO mg/L	7.80	7.08	19.61	6.70	7.01	7.06	8.40	7.85	127.2	7.80
Surf. pH	7.86	7.89	27.01	7.85	7.90	7.94	8.04	8.01	28.31	8.03
Bottom Temp °C	28.67	28.05	27.27	27.27	27.37	27.56	27.27	28.45	8.10	28.44
Bottom Salinity	25.70	31.24	27.77	27.40	28.48	26.83	24.77	24.69	28.19	28.47
Bottom DO%	193.4	230.3	121.9	115.3	136.7	107.7	104.5	125.5	126.4	163.2
Bottom DO mg/L	12.99	15.00	8.31	7.93	9.22	7.32	7.21	8.37	8.43	11.07
Bottom pH	8.40	8.45	8.03	8.01	8.15	7.96	7.93	8.07	8.11	8.30
NO_3^- (Nitrate) mg/L	0.00	0.06	0.00	0.00	0.26	0.48	0.09	0.00	0.00	0.00
NO_2^- (Nitrite) mg/L	0.004	0.003	0.003	0.005	0.001	0.005	0.004	0.010	0.000	0.004
NH_4 (Ammonia) mg/L	0.04	0.02	0.04	0.03	0.01	0.02	0.02	0.02	0.02	0.01
PO_4^{-3} (Phosphate)	+++	+++	+++	0.00	+++	+++	0.24	0.26	+++	1.12
SiO_2 (Silica) mg/L	8	11	11	8	4	4	6	0	3	8
Turbidity (FAU)	2	0	7	6	4	0	4	5	0	3

Water Quality Sampling Data Sheet:

Date: 11/6/14
Fishpond: He'e19

Initial Observations:

Start Time	8:15am	% Cloud Cover	70%
Finish Time	11/7 03:00pm		
Tide	rising	Precipitation	∅
Air Temp	80° F	Other Comments:	
Wind Speed / Direction	ENE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	52	65	66	50	78	57	54	55	62	64
Surf. Water Temp °C	24.48	23.63	23.98	23.57	23.3	24.29	24.62	25.36	24.81	24.97
Surf. Salinity	27.21	17.17	6.55	18.39	15.13	27.10	23.35	31.05	29.39	29.42
Surf. DO %	87.7	92.1	58.6	95.2	83.6	88.0	85.5	78.5	87.4	79.2
Surf. DO mg/L	6.26	7.06	4.84	7.46	6.31	6.3	6.23	5.37	6.12	5.53
Surf. pH	7.89	7.79	7.34	7.78	7.72	7.90	7.80	7.84	7.91	7.91
Bottom Temp °C	25.15	25.86	24.96	24.64	26.27	24.50	25.99	26.07	26.13	25.66
Bottom Salinity	27.66	20.51	24.12	22.04	33.32	27.18	28.66	32.37	33.27	31.89
Bottom DO%	75.5	80.9	76.4	84.4	42.0	82.9	75.1	69.4	69.3	67.3
Bottom DO mg/L	5.34	6.05	5.45	6.20	2.74	5.50	5.18	4.73	4.61	4.57
Bottom pH	7.86	7.76	7.72	7.78	7.71	7.83	7.77	7.82	7.88	7.86
NO_3^- (Nitrate) mg/L	0.36	0.00	0.00	0.53	0.29	0.70	0.84	0.67	0.53	0.37
NO_2^- (Nitrite) mg/L	0.009	0.007	0.000	0.005	0.004	0.002	0.000	0.002	0.006	0.007
NH_4 (Ammonia) mg/L	0.00	0.00	0.15	0.27	0.06	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.47	0.27	0.29	0.61	1.00	+++	+++	+++	+++	+++
SiO_2 (Silica) mg/L	5	8	9	18	17	5	7	4	6	3
Turbidity (FAU)	7	5	14	18	17	7	0	0	0	8

Water Quality Sampling Data Sheet:

Date: 11/11/2014
Fishpond: He'eia
Kealii & Kinthi

Initial Observations:

Start Time	12:00pm	% Cloud Cover	25%
Finish Time	3:00pm		
Tide	0.4m	Precipitation	none
Air Temp	27°C	Other Comments:	
Wind Speed/Direction	4 kts NNE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	57	67	68	62	98	62	54	55	72	62
Surf. Water Temp °C	26.10	26.27	26.72	26.61	26.53	26.78	27.29	27.56	27.30	26.55
Surf. Salinity	26.22	30.37	33.01	32.51	32.70	33.31	33.95	31.02	33.59	29.94
Surf. DO %	90.0	61.0	74.0	58.0	75.1	90.8	83.2	114.0	112.2	91.6
Surf. DO mg/L	6.27	4.51	5.51	3.87	4.93	6.03	5.43	7.43	7.33	6.18
Surf. pH	7.77	7.76	7.89	7.77	7.87	7.95	7.92	8.03	8.05	7.95
Bottom Temp °C	27.08	26.54	26.70	27.08	26.73	27.10	27.25	27.61	27.31	27.12
Bottom Salinity	32.80	31.17	33.44	33.61	34.51	34.33	33.97	31.01	33.62	33.89
Bottom DO%	85.7	97.0	61.0	79.7	107.0	107.2	76.6	106.1	112.3	109.7
Bottom DO mg/L	5.67	6.38	3.77	5.24	7.05	7.06	5.04	6.91	7.38	6.63
Bottom pH	7.94	8.01	7.91	7.89	8.01	8.05	7.90	8.03	8.06	8.03
NO_3^- (Nitrate) mg/L	1.67	1.04	0.26	0.39	0.50	1.12	1.81	1.44	2.57	0.88
NO_2^- (Nitrite) mg/L	0.007	0.002	0.006	0.004	0.005	0.001	0.000	0.002	0.005	0.001
NH_4 (Ammonia) mg/L	0.03	0.00	0.02	0.00	0.00	0.01	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	+++	+++	+++	+++	+++	1.98	+++	+++	+++	+++
SiO_2 (Silica) mg/L	9	8	9	4	5	4	3	2	4	8
Turbidity (FAU)	4	12	20	12	4	4	18	9	11	3

Water Quality Sampling Data Sheet:

Date 11/12/14
Location Hc'e-1a
Initial Observations:

Start Time	1:30pm	% Cloud Cover	25%
Finish Time	8:30am		
Tide	2.8m	Precipitation	None
Air Temp	27°C	Other Comments:	
Wind Speed / Direction	2kts		

Tested
11/14/14
06:30 = 08:30am
Kinda by feels

Site #	63	2	3	4	5	6	7	8	9	10
Depth (cm)	2840	61	65	64	81	52	52	51	63	61
Surf. Water Temp °C	23.80	28.22	27.83	27.93	28.13	28.35	28.23	29.75	29.31	28.54
Surf. Salinity	914	29.18	30.02	31.39	28.83	32.04	31.74	32.88	32.16	27.59
Surf. DO %	6.21	75.8	6.65	62.1	81.7	72.1	57.1	108.4	108.6	86.3
Surf. DO mg/L	7.83	5.03	4.41	4.08	5.45	4.71	3.81	6.90	7.02	5.75
Surf. pH	29.72	7.86	7.82	7.77	7.86	7.85	7.78	8.00	8.07	7.88
Bottom Temp °C	31.05	28.54	28.38	28.61	28.37	29.14	28.51	30.34	29.70	29.94
Bottom Salinity	85.8	33.87	32.90	32.78	32.89	32.89	32.46	33.54	32.91	33.26
Bottom DO%	5.48	101.5	78.2	60.7	94.4	103.9	64.5	136.7	138.0	153.4
Bottom DO mg/L	7.98	6.48	5.05	3.90	6.11	6.54	4.18	11.59	8.72	8.26
Bottom pH	↓↓	8.10	7.87	7.78	8.05	8.05	7.83	8.24	8.16	8.20
NO_3^- (Nitrate) mg/L	0.78	0.61	0.55	0.80	0.73	.73	0.35	0.59	0.57	0.21
NO_2^- (Nitrite) mg/L	0.000	0.000	0.009	0.000	0.007	0.002	0.001	0.002	0.001	0.000
NH_4 (Ammonia) mg/L	0.04	0.22	0.00	0.16	0.10	0.00	0.00	0.05	0.00	0.00
PO_4^{-3} (Phosphate)	1.72	0.36	0.78	+++	0.53	0.55	1.24	0.67	0.68	0.71
SiO_2 (Silica) mg/L	15	6	10	5	4	4	3	3	5	5
Turbidity (FAU)	9	2	12	2	2	0	3	2	4	3

Water Quality Sampling Data Sheet:

Date: 11/12/14
Hebpond: Hc-19

Initial Observations:

Start Time	1:30pm	% Cloud Cover	25%
Finish Time	8:30am		
Tide	2.8m	Precipitation	None
Air Temp	27°C	Other Comments:	
Wind Speed/Direction	2 kts		

Tested
11/14/14
06:30 = 08:30am
Kinda by feahi

Site #	63	2	3	4	5	6	7	8	9	10
Depth (cm)	2840	61	65	64	81	52	52	51	63	61
Surf. Water Temp °C	23.80	28.22	27.83	27.93	28.13	28.35	28.23	29.75	29.31	28.54
Surf Salinity	9114	29.18	30.02	31.39	28.83	32.04	31.74	32.88	32.16	27.59
Surf. DO %	6.21	75.8	6.05	62.1	81.7	72.1	57.7	108.4	108.6	80.3
Surf DO mg/L	7.83	5.03	4.41	4.08	5.45	4.71	3.81	6.90	7.02	5.75
Surf. pH	29.72	7.86	7.82	7.77	7.86	7.85	7.78	8.00	8.07	7.88
Bottom Temp °C	31.05	28.54	28.38	28.61	28.37	29.14	28.51	30.34	29.70	29.44
Bottom Salinity	85.8	33.87	32.40	32.78	32.09	32.89	32.46	33.54	32.91	33.26
Bottom DO%	5.48	10.5	7.82	6.07	94.4	103.9	64.5	186.7	138.0	154.1
Bottom DO mg/L	7.98	6.48	5.05	3.90	6.11	6.54	4.18	11.59	8.72	8.26
Bottom pH	↓↓	8.10	7.87	7.78	8.05	8.05	7.83	8.24	8.16	8.20
NO_3^- (Nitrate) mg/L	0.78	0.61	0.55	0.80	0.73	.73	0.35	0.59	0.57	0.21
NO_2^- (Nitrite) mg/L	0.000	0.000	0.009	0.000	0.002	0.002	0.001	0.002	0.001	0.000
NH_4 (Ammonia) mg/L	0.04	0.22	0.00	0.16	0.10	0.00	0.00	0.05	0.00	0.00
PO_4^{-3} (Phosphate)	1.72	0.36	0.78	+++	0.53	0.55	1.24	0.61	0.68	0.71
SiO_2 (Silica) mg/L	15	6	10	5	4	4	3	3	5	5
Turbidity (FAU)	9	2	12	2	2	0	3	2	4	3

Water Quality Sampling Data Sheet:

Date: 11-14-14
Fishpond: Heeia
Initial Observations: Carly & Heilei

Start Time	11:13	% Cloud Cover	75%
Finish Time	12:00		
Tide	High going out	Precipitation	None
Air Temp	78°-80	Other Comments:	Samples in Freezer!
Wind Speed/Direction	light SSE then NE		

Tested by Kinihi & Keahi
11/18/14
12:30pm - 02:45pm

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	66	80	77	66	88	69	64	67	71	71
Surf. Water Temp °C	26.16	25.23	25.88	26.34	25.75	25.56	26.67	26.29	25.37	26.36
Surf. Salinity	29.64	28.42	28.75	23.08	27.33	26.23	27.99	29.00	27.03	30.83
Surf. DO %	59.7	61.0	58.0	60.7	69.1	86.8	60.2	71.9	82.1	55.5
Surf. DO mg/L	3.95	4.29	4.03	4.29	5.14	6.14	4.05	4.93	5.78	4.00
Surf. pH	7.72	7.70	7.74	7.52	7.79	7.83	7.67	7.76	7.82	7.78
Bottom Temp °C	27.43	26.83	27.54	27.35	26.97	26.80	27.37	26.29	25.62	27.01
Bottom Salinity	31.50	33.54	32.15	31.70	32.54	32.08	31.08	29.09	27.98	33.07
Bottom DO%	59.4	73.2	54.8	51.1	85.7	68.9	48.3	74.1	79.8	113.8
Bottom DO mg/L	3.91	4.84	3.65	3.39	5.72	4.62	3.13	5.05	5.56	7.57
Bottom pH	7.73	7.89	7.72	7.69	7.91	7.83	7.71	7.79	7.81	8.07
NO_3^- (Nitrate) mg/L	0.00	0.00	0.00	0.69	0.00	0.60	0.00	0.00	0.08	0.52
NO_2^- (Nitrite) mg/L	0.014	0.006	0.003	0.010	0.000	0.006	0.003	0.000	0.003	0.003
NH_4 (Ammonia) mg/L	0.03	0.00	0.02	0.03	0.06	0.00	0.04	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.08	0.59	0.37	0.05	0.00	0.09	0.08	0.06	0.13	0.06
SiO_2 (Silica) mg/L	4	3	4	5	2	4	5	2	3	7
Turbidity (FAU)	0	2	26	7	7	0	8	7	4	3

Water Quality Sampling Data Sheet:

Date: 11/19/14
Fishpond: Hc'e19

collected by Kinohi

tested by Kinohi & Keahi 11/20/14 07.00 am - 9:[illegible] pm

Initial Observations:

Start Time	12.30pm	% Cloud Cover	75%
Finish Time	02:30pm		
Tide		Precipitation	none
Air Temp		Other Comments:	windy, sunny
Wind Speed/Direction	Mololani/ TRADES		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	57	71	70	52	110	64	56	56	60	67
Surf. Water Temp °C	24.20	24.75	24.93	24.97	24.46	25.26	25.30	25.63	25.45	25.31
Surf. Salinity	26.18	15.59	9.85	14.31	13.42	26.46	19.24	22.05	26.33	23.58
Surf. DO %	131.7	123.2	104.9	116.1	110.5	147.2	129.8	133.1	143.9	149.7
Surf. DO mg/L	9.69	9.34	8.20	8.84	8.46	10.51	9.59	9.64	10.16	10.85
Surf. pH	7.91	7.92	7.58	7.75	7.76	8.15	7.99	8.06	8.17	8.17
Bottom Temp °C	24.73	24.90	25.01	25.44	24.07	25.52	25.62	26.21	26.25	25.34
Bottom Salinity	23.67	24.14	21.74	17.69	23.54	28.72	19.09	26.48	28.52	24.37
Bottom DO%	177.9	140.9	134.4	124.0	136.9	167.5	143.7	183.4	169.2	159.6
Bottom DO mg/L	12.80	10.57	9.89	9.19	9.97	11.66	10.59	11.60	11.41	11.61
Bottom pH	8.21	8.14	7.87	7.91	8.08	8.18	8.02	8.21	8.22	8.21
NO_3^- (Nitrate) mg/L	0.20	0.00	0.06	0.00	0.00	0.35	0.14	0.00	0.88	0.00
NO_2^- (Nitrite) mg/L	0.000	0.000	0.001	0.000	0.006	0.000	0.001	0.000	0.006	0.000
NH_4 (Ammonia) mg/L	0.00	---	0.00	0.00	0.08	---	---	0.00	---	0.00
PO_4^{-3} (Phosphate)	0.67	1.41	0.55	0.66	1.39	+++	1.08	1.15	+++	+++
SiO_2 (Silica) mg/L	10	12	16	12	13	5	9	6	5	6
Turbidity (FAU)	8	12	10	9	13	0	8	10	8	6

Water Quality Sampling Data Sheet:

Date: 11/24/14
Fishpond: He'e ia

Kiholo Restricted
11/25/14
2:00pm - 04:30pm

Initial Observations:

Start Time	1:00 pm	% Cloud Cover	100%
Finish Time	2:12 pm		
Tide	Dropping	Precipitation	OFF and on all day
Air Temp		Other Comments:	
Wind Speed/Direction	NE		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	51	65	67	60	84	54	49	51	68	63
Surf. Water Temp °C	24.62	24.92	25.24	25.64	25.10	25.19	25.73	25.61	25.10	24.90
Surf. Salinity	30.12	31.60	31.33	32.85	31.71	32.40	32.80	32.95	31.72	31.63
Surf. DO %	106.7	109.1	99.1	89.2	103.5	116.2	87.4	96.3	115.3	112.5
Surf. DO mg/L	7.48	7.43	6.88	5.97	7.10	7.90	5.87	6.69	7.85	7.55
Surf. pH	7.86	7.92	7.88	7.90	7.92	7.96	7.87	7.91	7.95	7.97
Bottom Temp °C	24.76	25.00	26.01	26.45	25.61	25.19	26.19	25.63	25.14	24.91
Bottom Salinity	30.31	31.64	32.77	34.05	32.53	32.42	33.90	33.00	32.84	31.68
Bottom DO%	105.7	104.3	89.7	87.1	105.7	113.3	81.7	98.9	118.5	101.6
Bottom DO mg/L	7.40	7.20	6.05	5.78	7.22	7.80	5.45	6.72	8.13	7.13
Bottom pH	7.87	7.93	7.87	7.90	7.95	7.96	7.81	7.91	8.00	7.97
NO_3^- (Nitrate) mg/L	0.00	0.69	0.58	0.07	0.54	0.00	0.36	0.86	0.00	0.74
NO_2^- (Nitrite) mg/L	0.000	0.004	0.004	0.005	0.006	0.002	0.009	0.000	0.005	0.004
NH_4 (Ammonia) mg/L	0.03	0.02	0.01	0.00	0.00	0.00	0.04	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	0.40	0.39	0.11	+++	2.73	+++	0.23	0.14	+++	0.19
SiO_2 (Silica) mg/L	5	3	3	4	3	2	3	3	3	3
Turbidity (FAU)	9	3	4	6	15	5	7	4	0	0

Water Quality Sampling Data Sheet:

collected but not tested

Date: 11/28/14
Fishpond: Heeia

Initial Observations:

Start Time	06:30am	% Cloud Cover	70%
Finish Time			
Tide	Height missing	Precipitation	none
Air Temp		Other Comments:	constant wind, sunrise
Wind Speed/Direction			

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	80	92	83	76	140	83	80	79	84	82
Surf. Water Temp °C	20.73	20.80	20.98	20.77	21.04	22.90	20.85	22.11	20.45	20.83
Surf. Salinity	25.27	22.72	13.45	25.53	27.50	35.12	27.25	34.48	35.32	31.25
Surf. DO %	107.9	107.7	93.6	100.1	104.2	101.9	100.7	99.4	104.9	108.6
Surf. DO mg/L	8.17	8.42	7.83	7.74	7.87	7.14	7.63	7.07	7.28	7.88
Surf. pH	7.86	7.76	7.44	7.74	7.81	7.95	7.81	7.89	7.97	7.83
Bottom Temp °C	21.09	21.07	21.18	21.08	21.57	22.85	21.61	22.23	23.32	20.83
Bottom Salinity	31.98	32.42	30.03	30.07	31.33	35.11	31.95	34.56	35.29	31.22
Bottom DO%	88.2	78.7	75.1	81.7	94.1	92.3	86.9	95.5	107.3	93.9
Bottom DO mg/L	6.14	6.02	5.53	6.12	6.86	6.43	6.27	6.85	7.44	6.97
Bottom pH	7.85	7.81	7.58	7.73	7.83	7.94	7.78	7.88	7.97	7.83
NO_3^- (Nitrate) mg/L										
NO_2^- (Nitrite) mg/L										
NH_4 (Ammonia) mg/L										
PO_4^{-3} (Phosphate)										
SiO_2 (Silica) mg/L										
Turbidity (FAU)										

Water Quality Sampling Data Sheet:

Date: 11/3/14
Fishpond: Helia

Initial Observations:

Start Time		% Cloud Cover	80%
Finish Time			
Tide	OHm	Precipitation	Light Rain early Morning
Air Temp	27°C 30°C	Other Comments:	
Wind Speed/Direction	11 kts E 2pm		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	130	85	84	63	100	77	70	72	90	
Surf. Water Temp °C	26.74	26.82	27.14	27.57	26.97	26.79	27.33	27.38	27.04	27.42
Surf. Salinity	23.72	14.43	7.76	12.61	16.40	22.57	22.07	26.25	27.08	25.71
Surf. DO %	101.21	97.3	86.0	106.2	102.6	103.1	111.3	109.3	101.01	117.4
Surf. DO mg/L	7.89	7.15	6.55	7.81	7.46	7.24	7.77	7.46	7.37	8.03
Surf. pH	7.89	7.82	7.45	7.82	7.90	7.96	7.97	7.99	8.37	8.05
Bottom Temp °C	26.78	26.50	26.89	27.13	27.29	27.03	27.77	27.60	27.30	27.30
Bottom Salinity	29.01	27.81	26.09	27.76	30.81	32.43	32.10	32.04	33.58	33.14
Bottom DO%	135.7	112.1	106.2	113.5	132.4	115.6	116.6	110.2	162.0	137.0
Bottom DO mg/L	9.01	7.69	7.33	7.69	8.82	7.71	7.67	7.31	10.56	9.07
Bottom pH	8.13	7.99	7.92	7.93	8.09	8.03	8.02	8.03	8.16	8.14
NO_3^- (Nitrate) mg/L	1.25	0.29	0.36	0.33	0.43	0.85	0.93	0.69	0.00	0.42
NO_2^- (Nitrite) mg/L	0.000	0.003	0.000	0.000	0.003	0.001	0.003	0.003	0.000	0.001
NH_4 (Ammonia) mg/L	0.00	0.03	0.02	0.03	0.03	0.03	0.02	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	2.14	0.45	1.51	1.83	+++	1.09	1.62	1.07	0.34	2.59
SiO_2 (Silica) mg/L	5	9	11	13	9	3	7	3	6	5
Turbidity (FAU)	0	4	11	0	3	0	0	0	0	0

Water Quality Sampling Data Sheet:

Date: 12/1/14
Fishpond: He'eia

12/3 7:00am – 9:00am

Initial Observations:

Start Time	1 00	% Cloud Cover	90%
Finish Time	2:15		
Tide 10:40 am:	1.62 ft Dropping	Precipitation	Passing Showers
Air Temp	25°C	Other Comments:	
Wind Speed/Direction	17 Kts		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	55	65	63	60	95	60	57	52	64	65
Surf. Water Temp °C	21.73	21.97	21.65	21.85	21.86	22.44	22.04	22.05	22.43	22.15
Surf. Salinity	26.74	29.86	29.15	25.14	25.02	29.93	27.13	26.80	27.93	28.62
Surf. DO %	132.0	129.5	95.8	123.6	120.6	138.0	131.8	136.0	143.0	147.0
Surf. DO mg/L	9.91	9.52	7.57	9.42	9.51	10.10	9.80	10.17	10.60	11.01
Surf. pH	7.91	7.97	7.87	7.89	7.84	8.00	7.97	7.98	8.06	8.07
Bottom Temp °C	21.76	22.36	21.97	22.12	22.37	22.49	22.05	22.06	23.33	22.18
Bottom Salinity	27.05	31.16	26.48	28.21	29.87	30.40	27.10	26.80	32.55	29.63
Bottom DO%	130.7	129.7	116.3	126.0	130.4	137.4	130.5	131.0	136.7	146.0
Bottom DO mg/L	9.81	9.41	8.66	9.45	9.54	10.04	9.77	9.79	12.85	10.78
Bottom pH	7.94	8.01	7.91	7.96	7.98	8.01	7.97	7.98	8.20	8.08
NO_3^- (Nitrate) mg/L	0.73	0.46	0.30	0.68	1.10	0.47	0.43	0.48	0.44	0.21
NO_2^- (Nitrite) mg/L	0.005	0.000	0.007	0.011	0.000	0.007	0.005	0.010	0.008	0.000
NH_4 (Ammonia) mg/L	0.00	---	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	+++	2.72	0.74	0.61	+++	+++	0.78	+++	0.33	+++
SiO_2 (Silica) mg/L	7	4	15	9	7	4	7	6	7	6
Turbidity (FAU)	1	12	15	5	3	3	0	0	0	0

Water Quality Sampling Data Sheet:

Date: 12/8
Fishpond: He·e·a
Initial Observations: kcan

Tested 12/9 - 7:00 am
kcan & kanahi-

Start Time	1:30	% Cloud Cover	50%
Finish Time	3:00		
Tide 2:16pm	1.30 ft	Precipitation	None
Air Temp	26°C	Other Comments:	
Wind Speed/Direction	2K 15 NNW		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	46	53	~~60~~ 30	55	69	52	46	50	70	57
Surf. Water Temp °C	27.43	27.06	27 ~~84~~	27.24	27.28	27.51	27.10	27.54	27.05	27.41
Surf. Salinity	34.71	28.25	27.51	31.53	32.95	32 ~~77~~	31.32	32.75	33.17	33 ~~79~~ 32
Surf. DO %	115.1	101.9	90.0	86.4	100.5	110.1	99.3	107.7	122.5	126.1
Surf. DO mg/L	7.51	6.03	6.11	~~5~~ 5.72	6.63	7.32	6.61	7 ~~22~~	8.11	8.40
Surf. pH	7.97	7.96	7.83	7.86	7.91	8.02	7.89	8.00	8.02	8.03
Bottom Temp °C	27.34	27.86	27.37	27.87	27.67	28.54	27.20	27.75	26.69	28.40
Bottom Salinity	31.82	34.46	34.15	34.36	34.59	34.53	31.86	32.98	33.82	34.39
Bottom DO%	112.5	166.1	82.3	106.7	12	168.9	94.6	120.6	122.5	152.9
Bottom DO mg/L	7.36	10.42	5.40	6.58	8.60	10.67	6.27	7.78	8.11	9.87
Bottom pH	7.98	8.15	7.80	7.92	8.00	8.01	7.91	8.02	8.04	8.13
NO_3^- (Nitrate) mg/L	0.10	0.00	0.00	0.00	0.13	0.25	0.00	0.35	0.50	0.00
NO_2^- (Nitrite) mg/L	0.002	0.000	0.000	0.000	0.003	0.006	0.005	0.009	0.003	0.000
NH_4 (Ammonia) mg/L	0.00	0.00	0.10	0.20	0.08	0.01	0.02	0.01	0.00	0.03
PO_4^{-3} (Phosphate)	0.81	1.65	+++	+++	+++	+++	+++	+++	+++	+++
SiO_2 (Silica) mg/L	3	13	9	6	4	9	3	3	2	5
Turbidity (FAU)	0	5	15	5	0	5	0	0	7	5

Water Quality Sampling Data Sheet:

Date: 12/07/14
Fishpond: He'eia

Initial Observations:

Start Time	12:45pm	% Cloud Cover	10% "Humid as"
Finish Time			
Tide	On rising	Precipitation	nine
Air Temp	26°C	Other Comments:	
Wind Speed/Direction	2 Knots WSW		

Site #	1	2	3	4	5	6	7	8	9	10
Depth (cm)	47	60	52	59	75	52	46	51	73	57
Surf. Water Temp °C	26.74	26.42	25.26	26.86	26.75	27.12	26.85	27.40	27.10	26.62
Surf. Salinity	20.11	13.10	30.80	33.10	31.84	34.13	33.81	34.97	34.98	34.24
Surf. DO %	95.0	85.2	59.2	79.2	101.9	97.5	79.1	115.6	132.9	87.9
Surf. DO mg/L	6.19	6.30	4.15	5.22	6.24	6.39	5.24	7.54	9.53	5.73
Surf. pH	7.86	7.42	7.55	7.18	7.79	7.91	7.83	7.99	8.04	7.88
Bottom Temp °C	27.33	27.01	26.79	27.07	27.09	27.70	27.13	27.50	27.12	27.19
Bottom Salinity	33.67	34.25	33.03	33.82	34.81	34.94	34.75	34.94	35.00	34.80
Bottom DO%	104.7	101.7	62.3	78.1	91.2	141.5	89.2	120.4	139.5	117.8
Bottom DO mg/L	6.84	7.11	4.10	5.14	5.85	9.41	5.84	7.82	8.89	7.82
Bottom pH	7.95	7.99	7.69	7.77	7.85	8.14	7.88	8.00	8.01	8.03
NO_3^- (Nitrate) mg/L	0.83	0.91	0.00	0.00	0.19	0.93	1.35	1.33	1.36	0.99
NO_2^- (Nitrite) mg/L	0.006	0.000	0.025	0.003	0.008	0.002	0.005	0.007	0.002	0.004
NH_4 (Ammonia) mg/L	0.01	0.03	0.03	0.00	0.00	0.00	0.00	0.00	0.00	0.00
PO_4^{-3} (Phosphate)	1.24	0.48	1.12	0.55	+++	+++	1.07	+++	0.62	+++
SiO_2 (Silica) mg/L	13	10	17	3	4	3	1	3	2	2
Turbidity (FAU)	4	14	22	0	0	0	9	0	0	0

→ .5 DI water .5 pond water / numbers not recalculated